

DUI2 - Dials User Interface

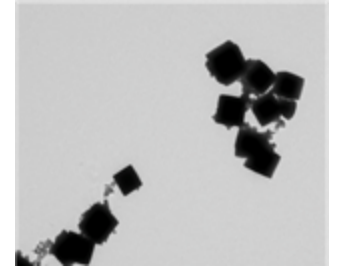
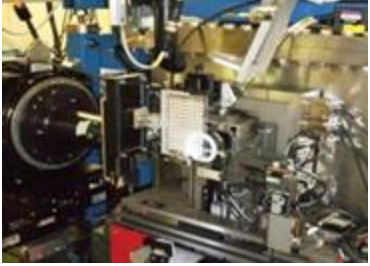
Dr. Luis Fuentes-Montero (Luiso)



What is DIALS?

Diffraction Integration for Advanced Light Sources

- Project began in late 2011 at Diamond Light Source and CCP4
- Additional funding by BioStruct-X and now Wellcome Trust
- Aim to develop new data processing software to meet modern challenges
- Now an international collaborative development

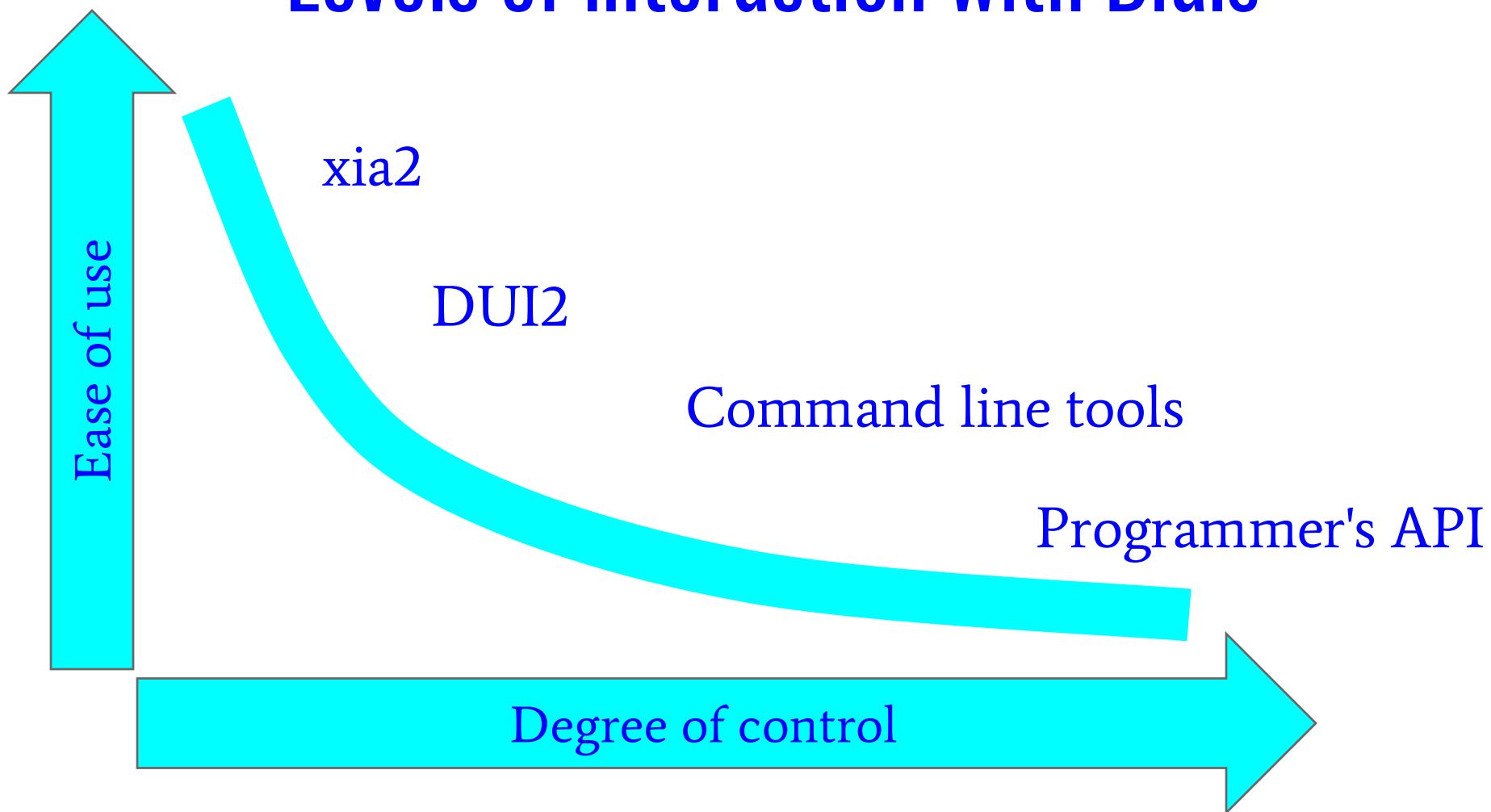


DIALS commands

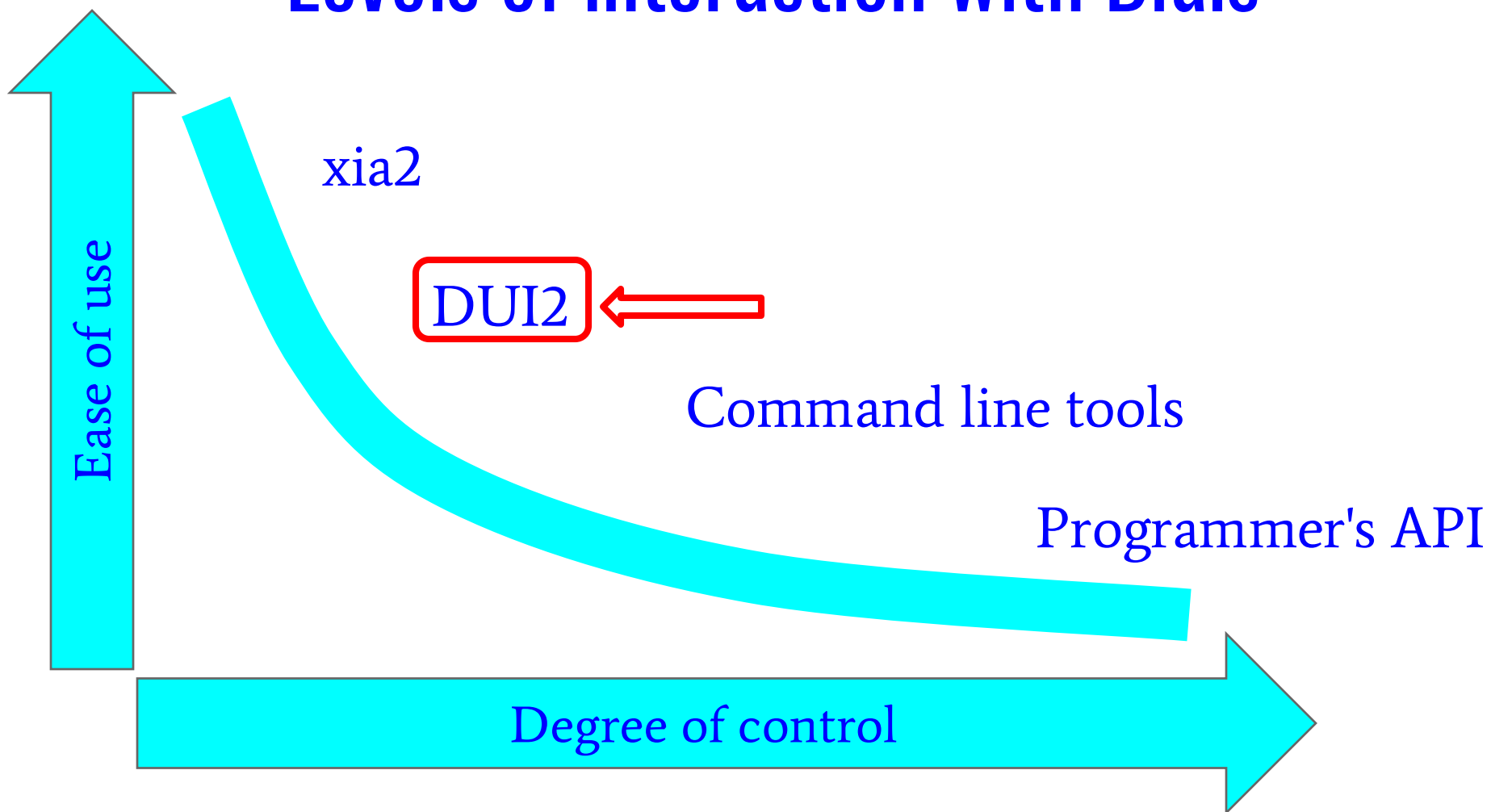
- `dials.import`
- `dials.find_spots`
- `dials.index`
- `dials.refine_bravais_settings`
- `dials.reindex`
- `dials.refine`
- `dials.integrate`
- `dials.symmetry`
- `dials.scale`
- `dials.export`

More than 90 `dials.*` commands

Levels of interaction with Dials



Levels of interaction with Dials



What is DUI?

What is DUI?

Drive Under Influence

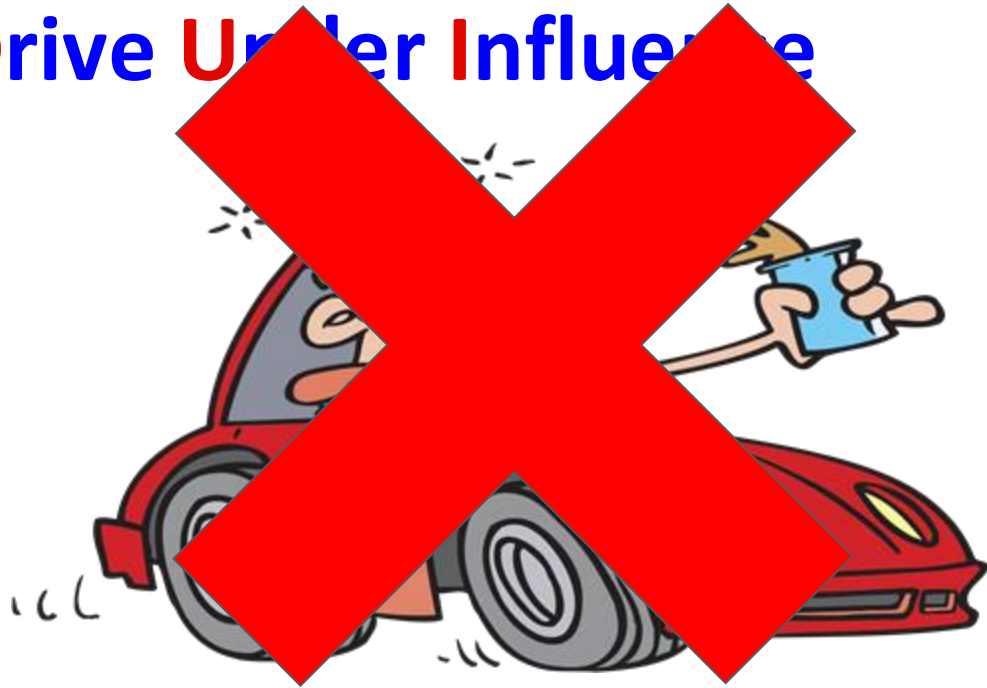
What is DUI?

Drive Under Influence



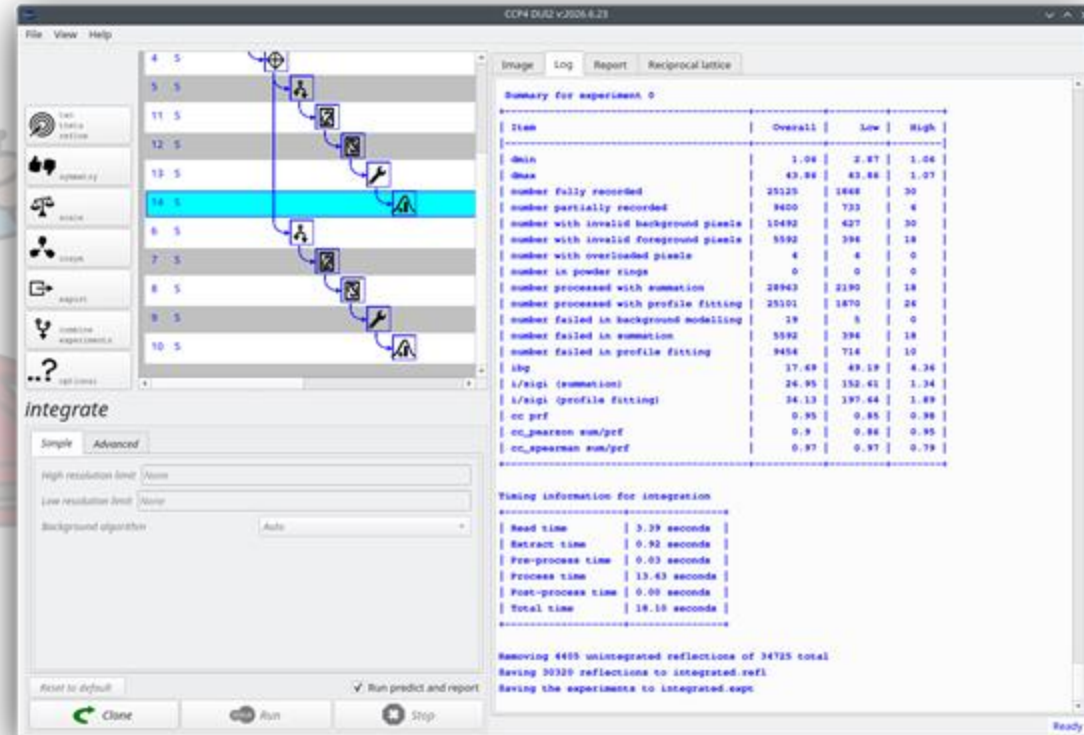
What is DUI?

Drive Under Influence



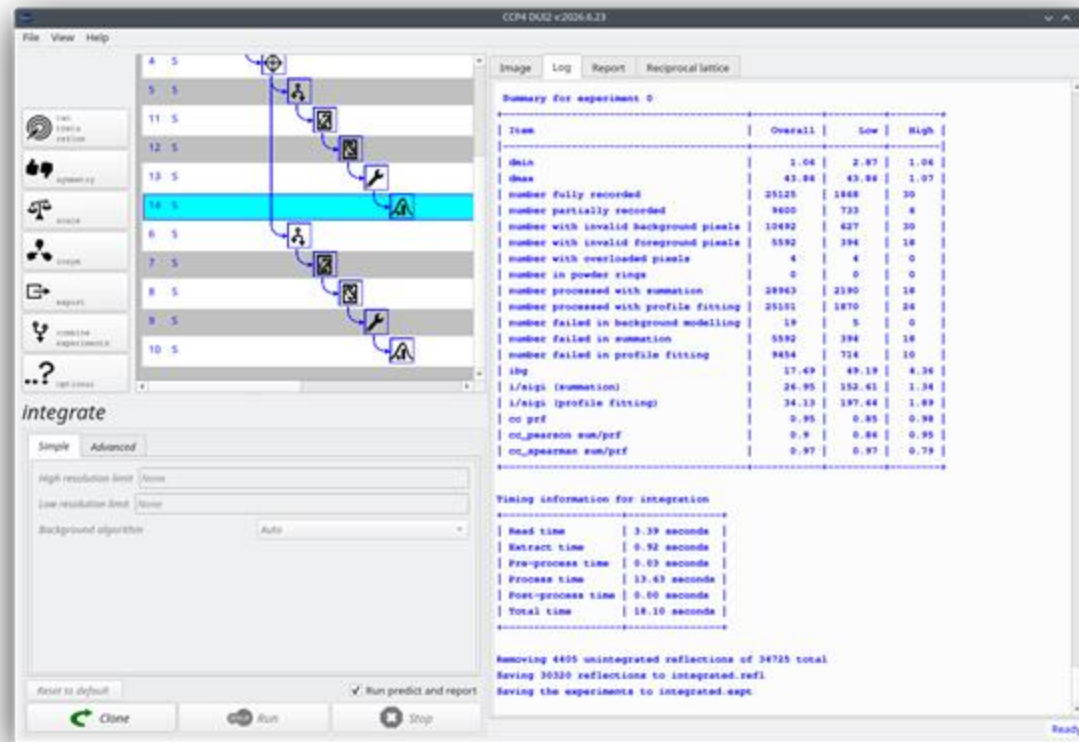
What is DUI?

Dials User Interface



What is DUI?

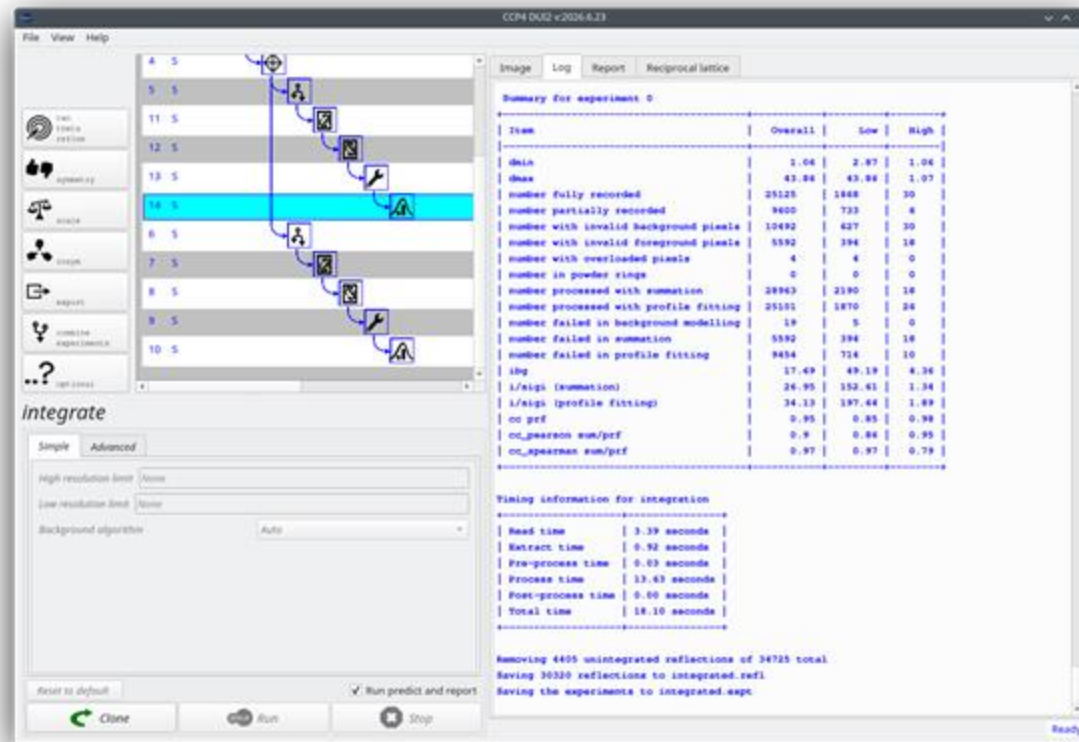
Dials User Interface



What is DUI?

Dials User Interface

Version 2



Why reach for Dui2?

Dui2 processes a single data set through DIALS with a graphical interface — an alternative to Xia2 for when you want direct, hands-on control over every DIALS command, rather than running from the command line.



Full command control

Step through import, find spots, index, refine, integrate and more — each one a DIALS command you configure and run yourself.



When Xia2 struggles

Users often turn to Dui2 when Xia2 hits problems or fails to complete — it lets you diagnose and steer processing directly.



A gentle on-ramp

This walkthrough introduces the basic steps using a simple example, before tackling more complex scenarios.

GETTING STARTED

Importing the dataset

Every Dui2 session begins the same way — bring your images in, then let DIALS build the geometry model.

1 Open images

Click “Open images” on the left panel.

2 Select a file

Choose any single image from your dataset.

3 Run

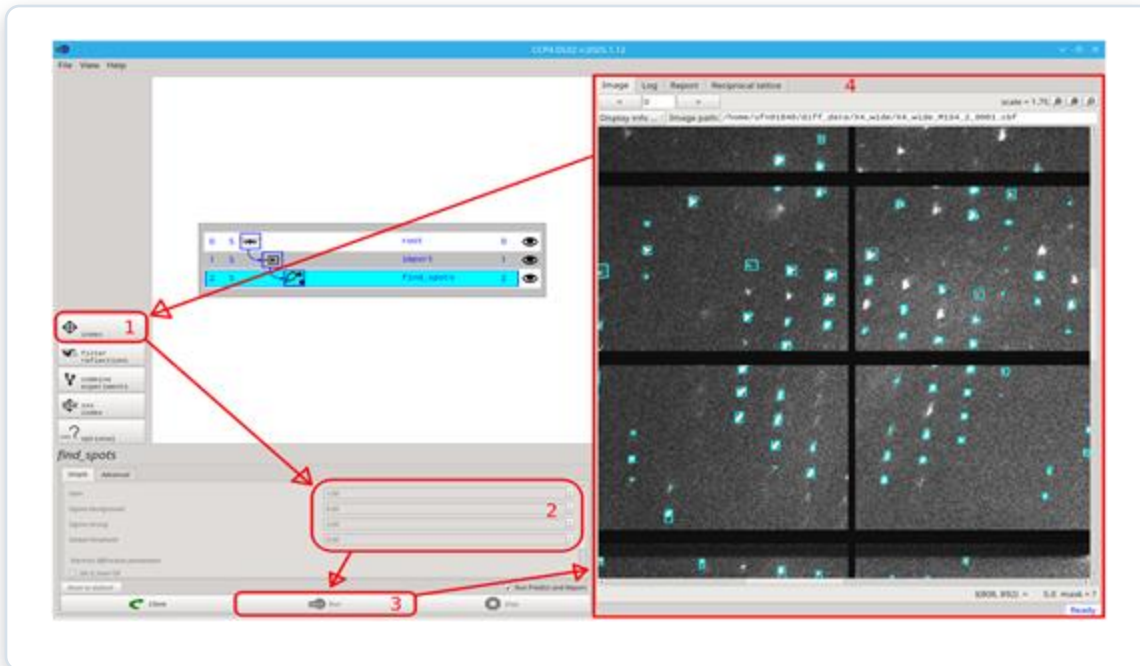
Click “Run” — the button with the DIALS logo.

This triggers the first DIALS command in the “import” node. Dui2 then walks you through the same four-step pattern for every subsequent command in the processing graph.

THE CORE PATTERN

The four-step processing loop

Once importing is done, every subsequent DIALS command in Dui2 follows the same repeating pattern — shown here for the “find spots” node.



The Dui2 interface, annotated with the four-step pattern.

1 Select a command

Pick a button in the command column (top left).

2 Adjust parameters

Edit values in the Simple or Advanced tabs.

3 Execute the command

Click “Run” and wait for it to finish.

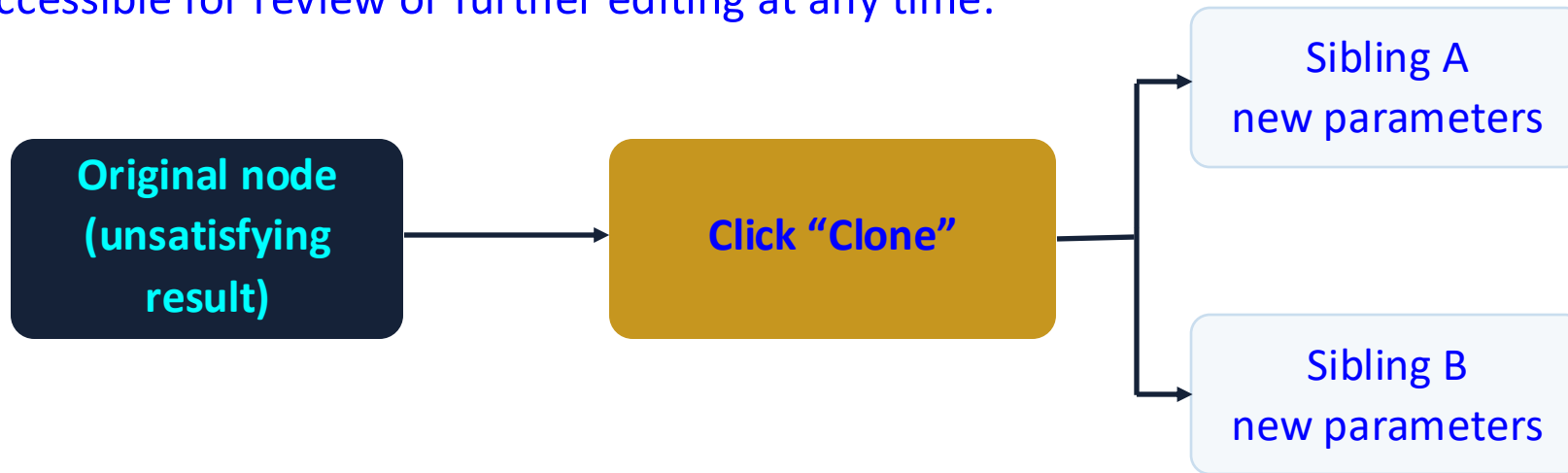
4 Review the output

Check the Log/Report/Image tabs, then repeat.

↻ loops back to step 1 until you're happy with the result

Branching with *Clone*

Not happy with a result? Fork the workflow instead of overwriting it — Dui2 keeps every branch and node accessible for review or further editing at any time.

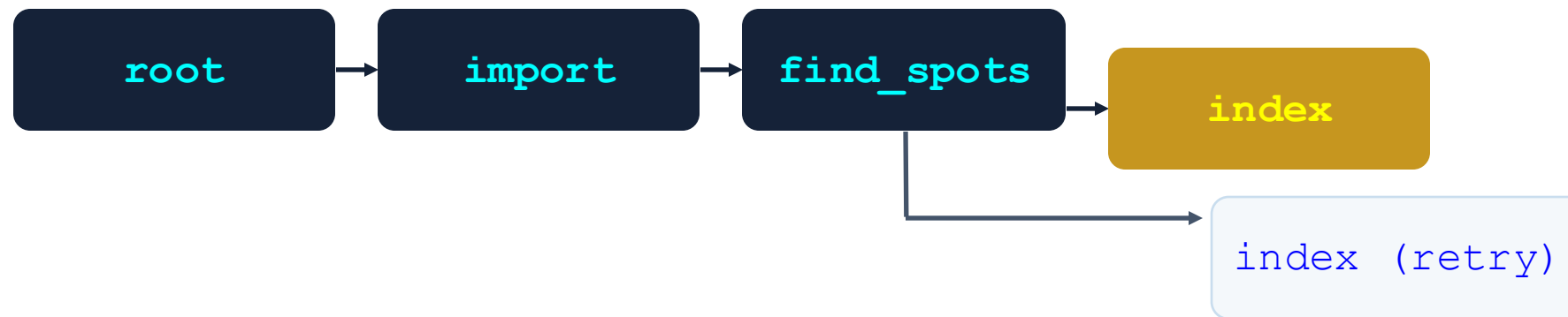


Modify the parameters and click "Run" again — Dui2 creates a sibling node so you can explore an alternative processing path without losing the original.

THE CORE IDEA

A tree of processing steps

Every DIALS command you run in Dui2 creates a new node in a tree. Click any node to jump back to that point and continue — or fork the tree to try a different path without losing what you already have.



Clicking “Clone” on an earlier node forks a sibling — a second attempt with different parameters, sitting right alongside the first.

The rest of this deck walks through a typical session, node by node.

Usage Workflow (ideal case)

A typical DUI2 session, node by node — from launching the app to a scaled, merged MTZ.

Import → Find spots

Bring in images, locate reflections

Index → Refine

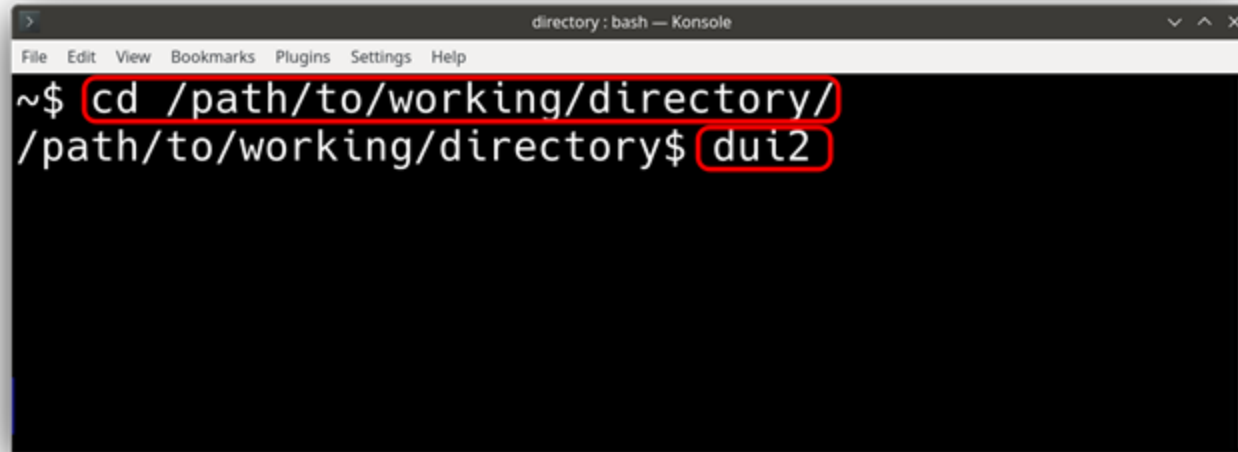
Model the lattice and geometry

Integrate → Scale

Measure and merge intensities

Launch Dui2

1 step 1 of 10



A screenshot of a terminal window titled "directory : bash — Konsole". The terminal shows the following commands and their output:

```
~$ cd /path/to/working/directory/  
/path/to/working/directory$ dui2
```

The commands are highlighted with red boxes.

Start Dui2 from inside a working directory. Dui2 writes its files (logs, .expt/.refl files, etc.) to the directory it was launched from, so cd into a dedicated processing directory first.

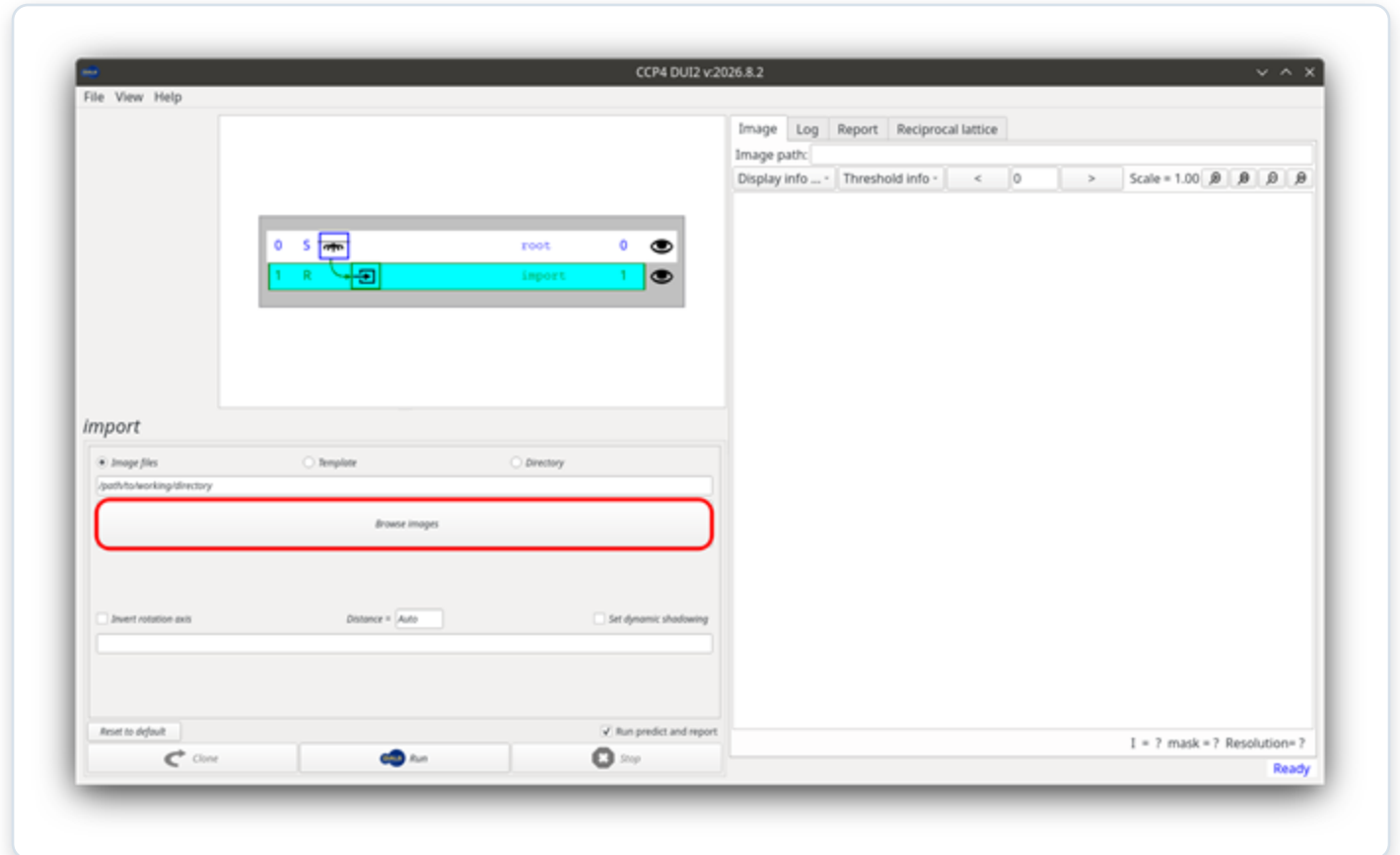
Import

2 step 2 of 10

Click “Open images”, navigate to your data, and select one image (or the master/.nxs file for HDF5/Eiger data). Dui2 builds a template matching the full data set, then “Run” reads the experiment metadata.

```
dials.import
```

Output: `imported.expt`



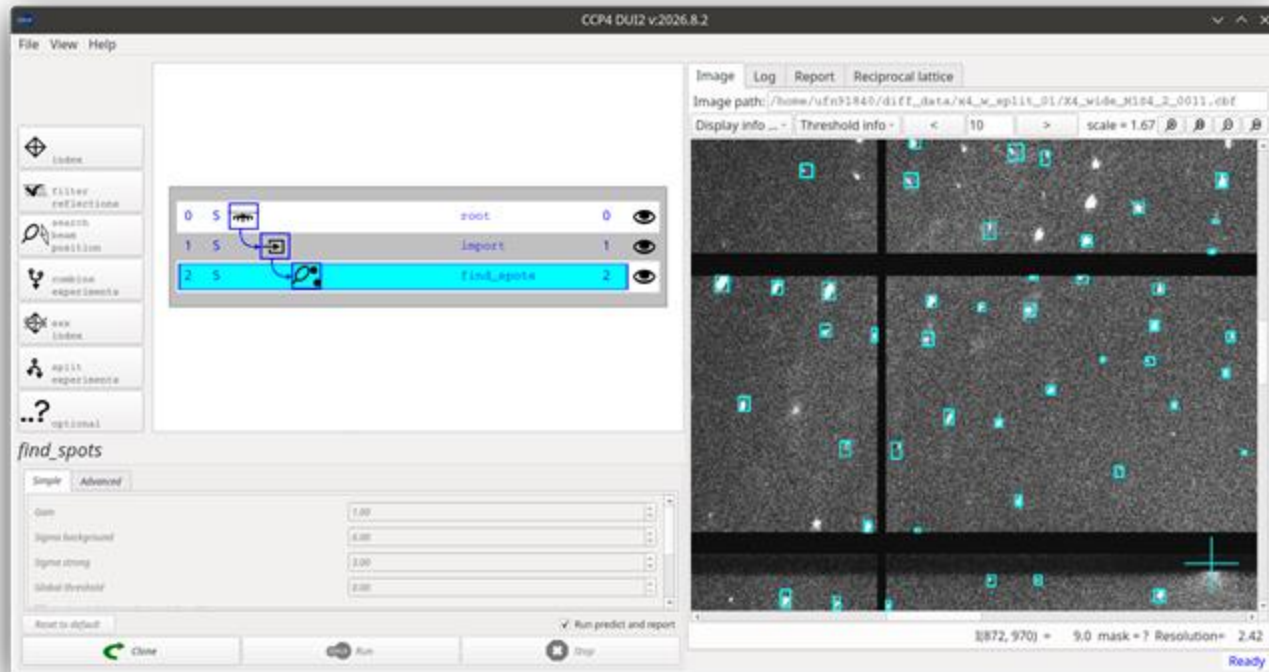
Find spots

3 step 3 of 10

Click the “find spots” button to create a new node, then “Run”. This searches every image for diffraction spots — one of the more time-consuming steps, since it reads the whole data set.

```
dials.find_spots
```

Output: strong.refl

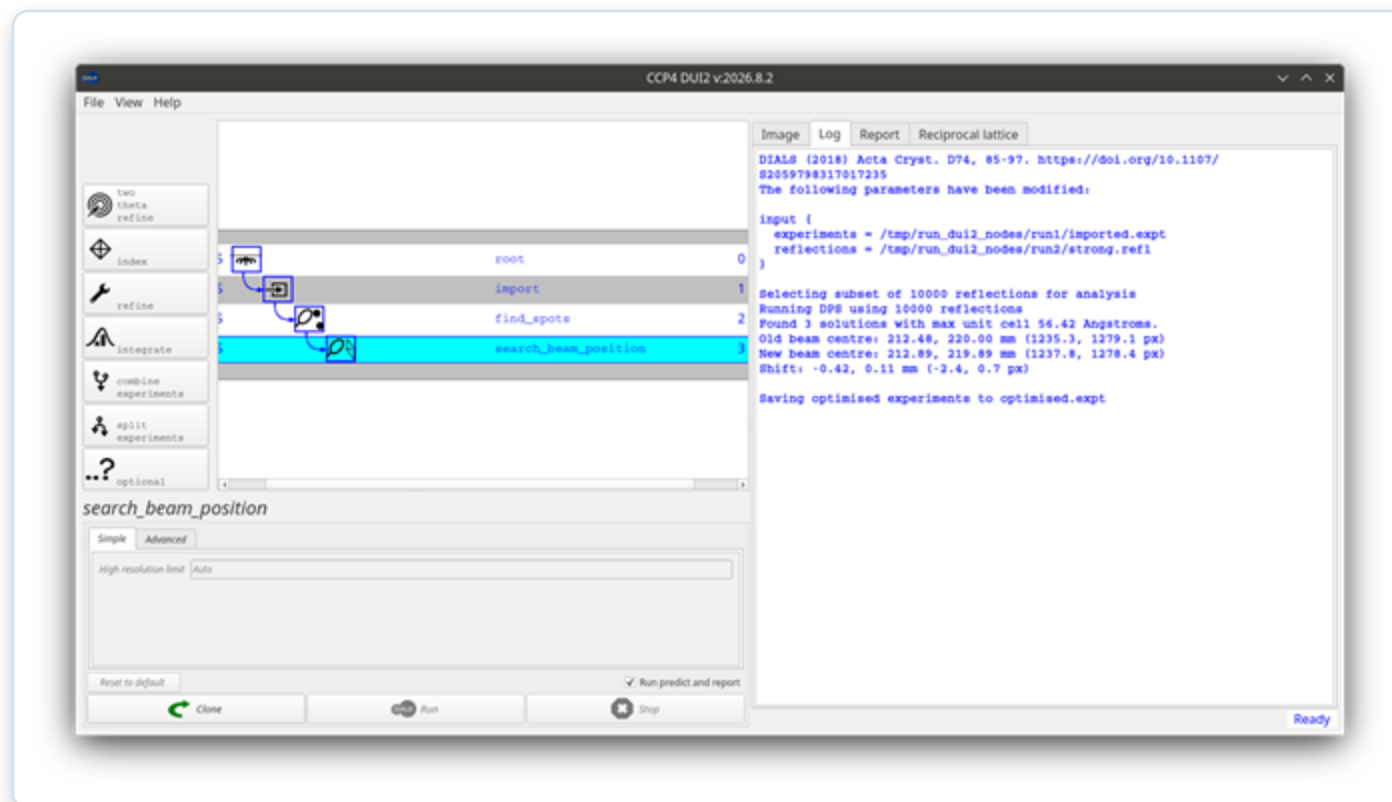


Search beam position

4 step 4 of 10

If the beam centre looks off in the Reciprocal Lattice viewer, this optional step refines it before indexing — worth a look if indexing struggles later on.

`dials.search_beam_position`



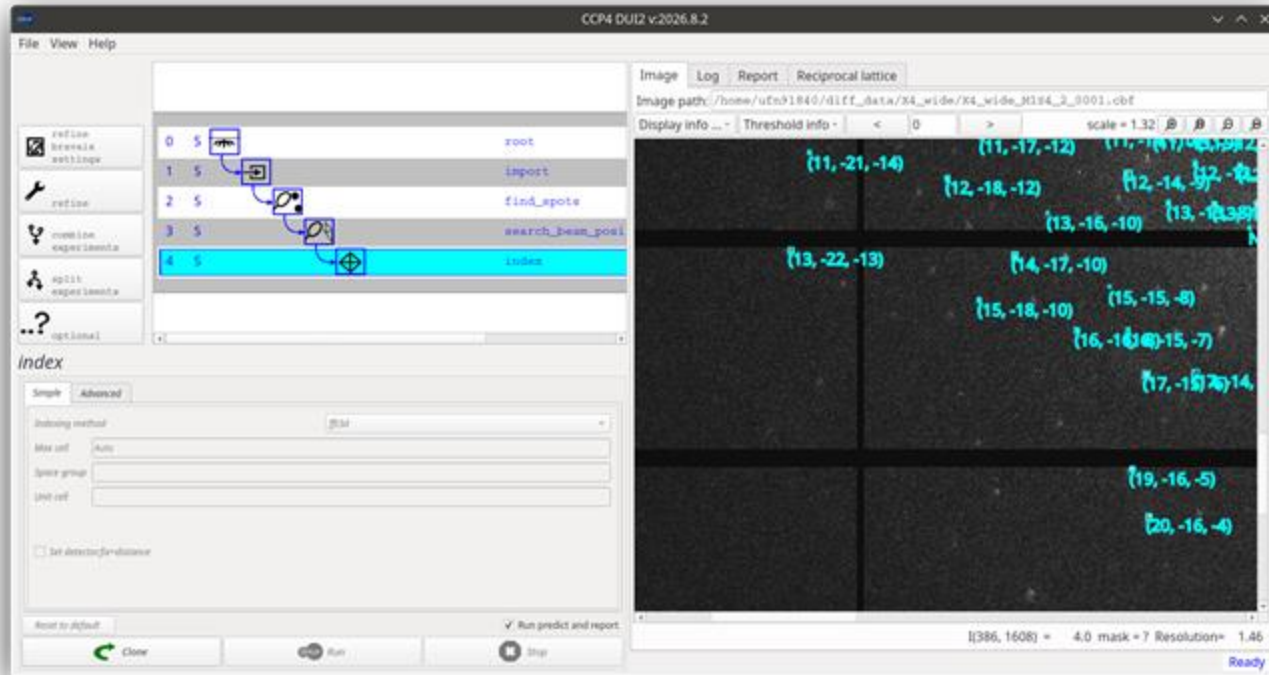
Index

5 step 5 of 10

Click “index” and “Run”. This finds the crystal lattice from the spot positions — the geometric heart of the whole processing pipeline.

`dials.index`

Output: `indexed.expt` /
`indexed.refl`

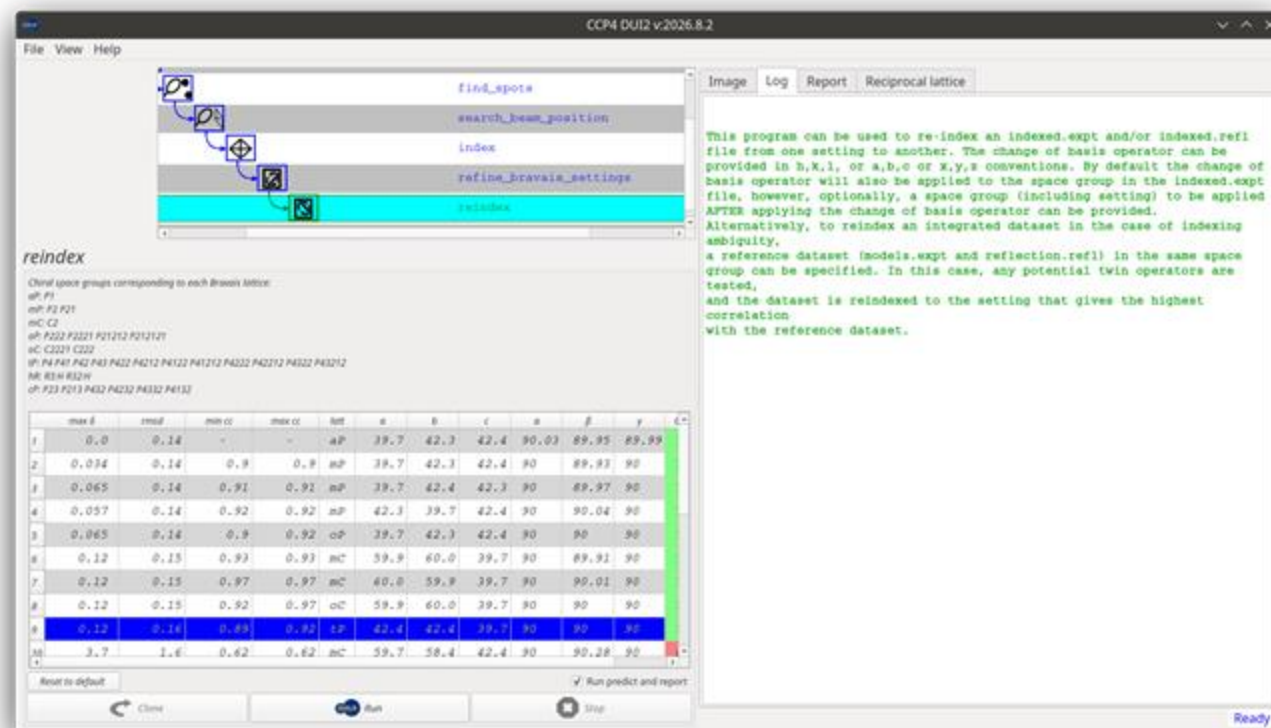


Determine the Bravais lattice

6 step 6 of 10

Proposes possible higher-symmetry lattices consistent with the indexed cell. If you pick one, a “Reindex” step applies it to the model.

dials.refine bravais settings

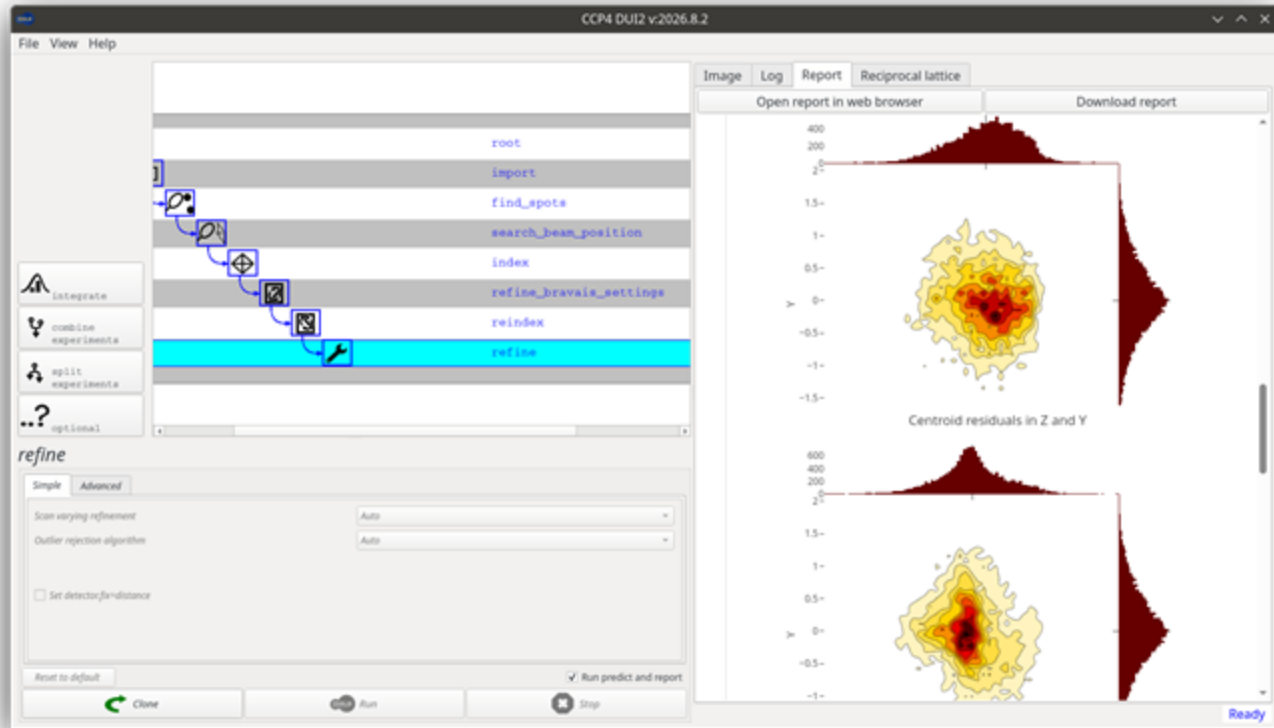


Refine

7 step 7 of 10

Click “Refine” to improve the crystal and experimental geometry model, including scan-varying refinement across the rotation range.

`dials.refine`

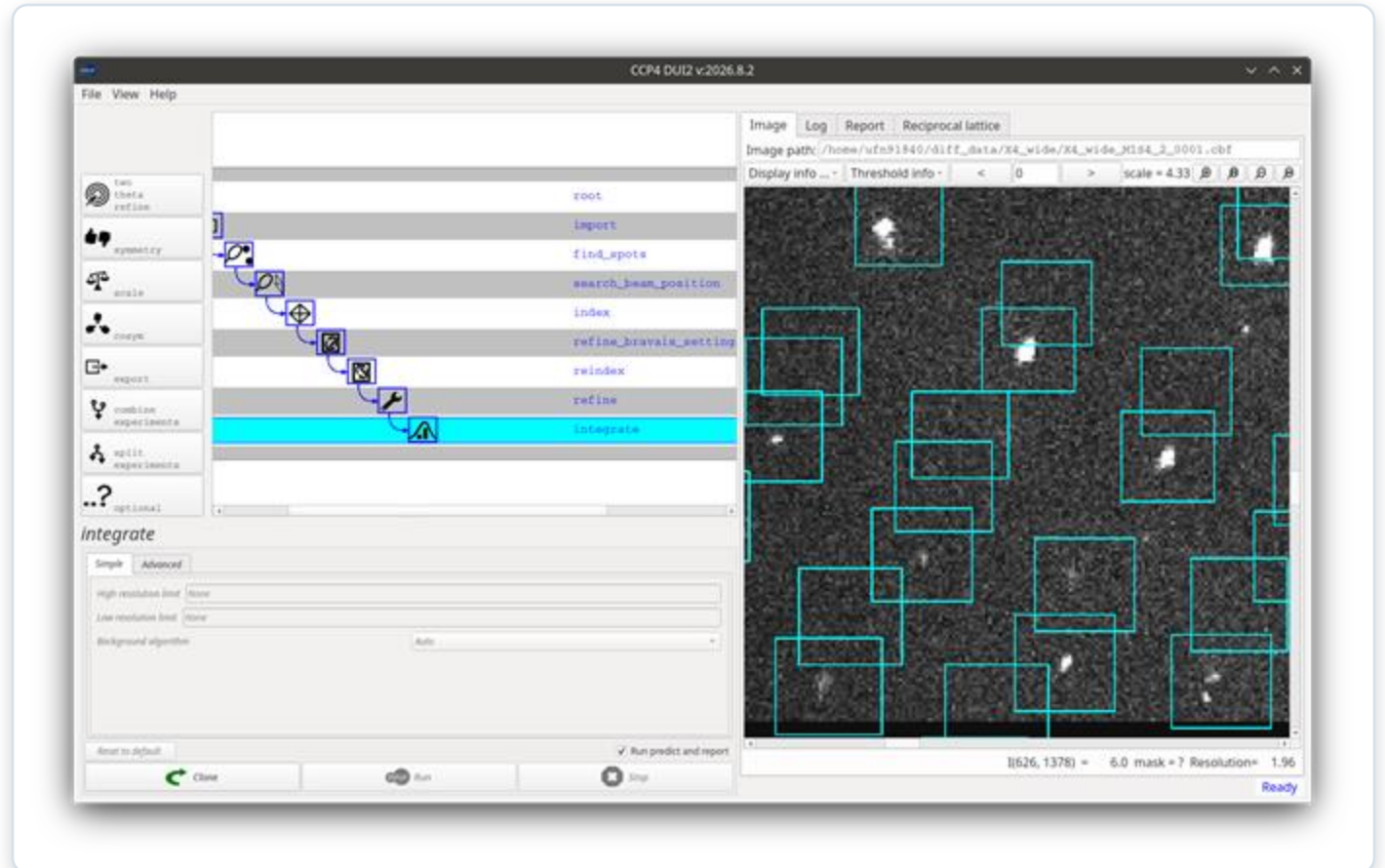


Integrate

8 step 8 of 10

Click “Integrate” and “Run” to measure the intensity of every predicted reflection. This is normally the most computationally expensive step in the whole workflow.

`dials.integrate`



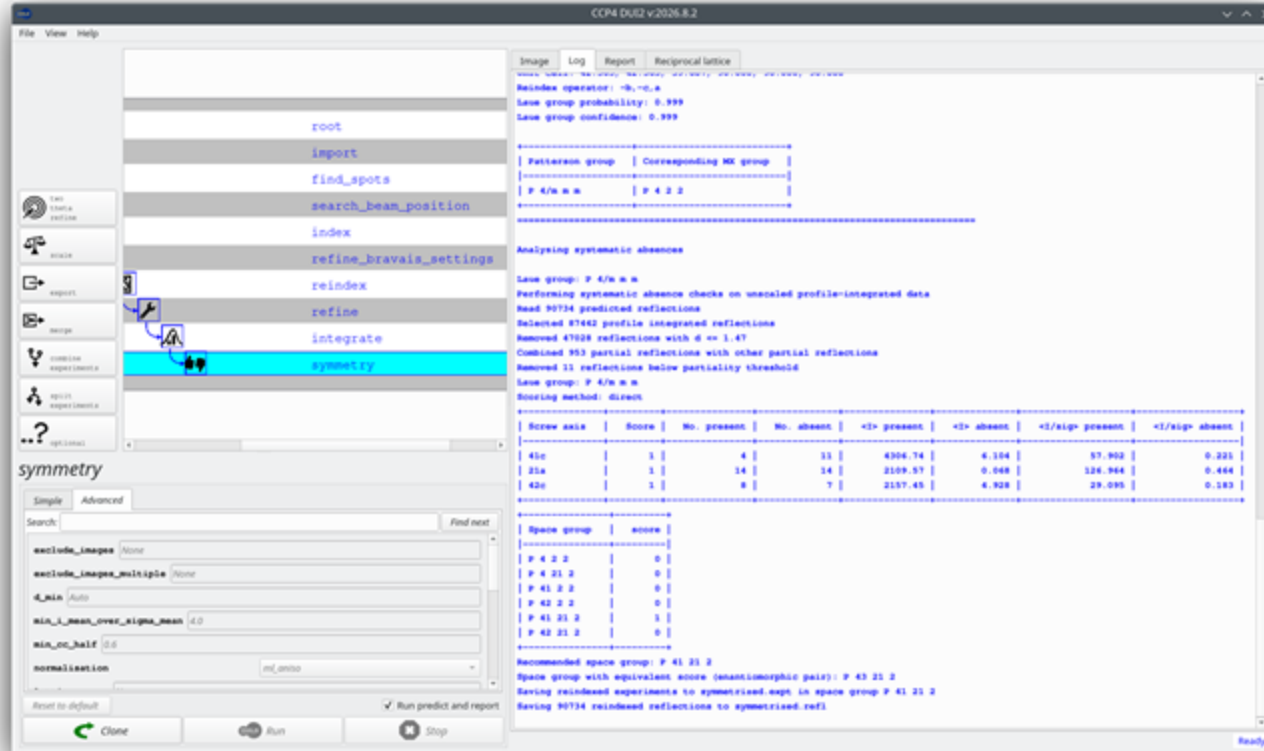
Symmetry

9 step 9 of 10

Click “symmetry” to determine the crystal's true symmetry, including a search for systematic absences that pin down screw axes.

dials.symmetry

Output: `symmetrized.expt` /
`symmetrized.refl`



Scale (and merge)

10 step 10 of 10

Click “scale” to correct for radiation damage, beam intensity variation, and absorption.

From here, merge for a merged MTZ, and/or export for a scaled, unmerged MTZ — both downloadable from their node.

`dials.scale` → `dials.merge` / `dials.export`

The screenshot displays the CCP4 DIALS v2026.8.2 interface. On the left, a workflow graph shows steps: root, import, find_spots, search_beam, index, refine_brav, reindex, refine, integrate, symmetry, and scale. The 'scale' node is highlighted in blue. Below the graph, the 'scale' panel is visible with 'Simple' and 'Advanced' tabs. The 'Simple' tab shows fields for 'High resolution limit' (None), 'Low resolution limit' (None), 'Model' (physical), and 'Error optimisation model' (basic). At the bottom of the panel are buttons for 'Reset to default', 'Clone', 'Run', and 'Stop'. On the right, the 'DIALS scaling report' is displayed, showing 'Merging statistics' and 'Error model details'. The 'Merging statistics' table is as follows:

	Overall	Low resolution	High resolution
Resolution (Å)	42.36 - 1.02	42.40 - 2.76	1.04 - 1.02
Observations	85380	6046	205
Unique reflections	16633	1074	172
Multiplicity	5.1	5.6	1.2
Completeness	87.24%	100.00%	18.49%
Mean I(σ)	9.6	68.3	-0.0
R _{merge}	0.088	0.036	-1.794
R _{meas}	0.097	0.039	-2.463
R _{pim}	0.040	0.056	-1.675
CC _{1/2}	0.999	0.999	-0.047

The 'Error model details' section shows 'Error model parameters' and 'Analysis plots'. The bottom right corner of the window shows a 'Ready' status.

Inspect, and iterate if needed

Log

The command's raw text output.

Report

An HTML `dials.report` summary.

Image

Diffraction images with overlays.

Reciprocal lattice

The reciprocal-space view of spots.

If a step doesn't go the way you want, click back on an earlier node, fork a new path with “Clone”, and try again with different parameters — without losing your earlier work.

For more detail

Legacy tutorial (full step-by-step walkthrough) • Multi-lattice • THAS1 (anomalous SAD phasing) • Beta lactamase

How to treat a *Multiple Lattices* case:

Full tutorial:

<https://github.com/ccp4/DUI2/wiki/Multiple-lattices>

This tutorial covers processing a dataset containing two overlapping crystal lattices using DIALS via DUI2. If you have not yet worked through a single-lattice dataset in DUI2, it is worth doing that first — this tutorial focuses on where the workflow diverges and why.

Dataset: semi-synthetic two-lattice trypsin dataset collected at beamline I04, Diamond Light Source (Zenodo record 10820).

Chapters:

Discover

Process as a single-lattice dataset first, use DUI2's visualisation tools to diagnose the second lattice, then re-index with `max_lattices=5` and split the history tree into two independent branches.

Process

Follow each branch through Bravais settings, refinement, and integration independently.

Post integration

Scale and export each lattice independently.

CHAPTER 1

Discover

Run single-lattice processing first, use DUI2's visualisation tools to confirm a second lattice, then re-index and split the tree.

Import images · Find spots · Index (single lattice) · Clone & re-index · Split experiments

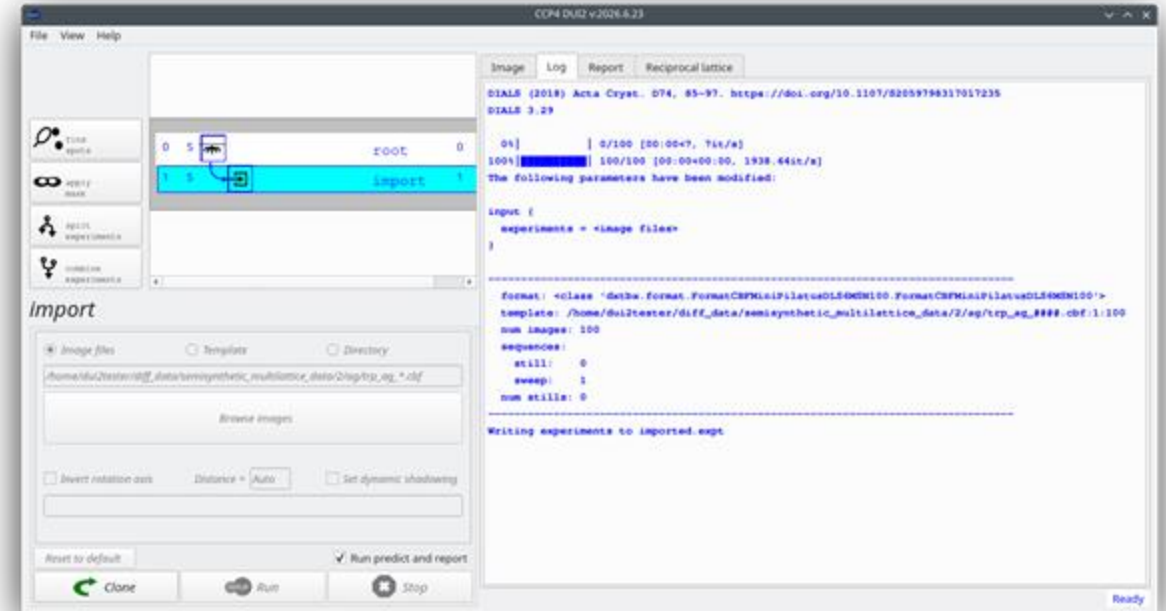
CHAPTER 1 · DISCOVER

Step 1

Import images

- When DUI2 starts, an incomplete `dials.import` node is already highlighted in the history tree.
- Click "Open images", navigate to the image directory, and select one CBF file, then click Run.

The log confirms 100 images interpreted as a single sweep, written to `imported.expt`.



The DUI window at start up

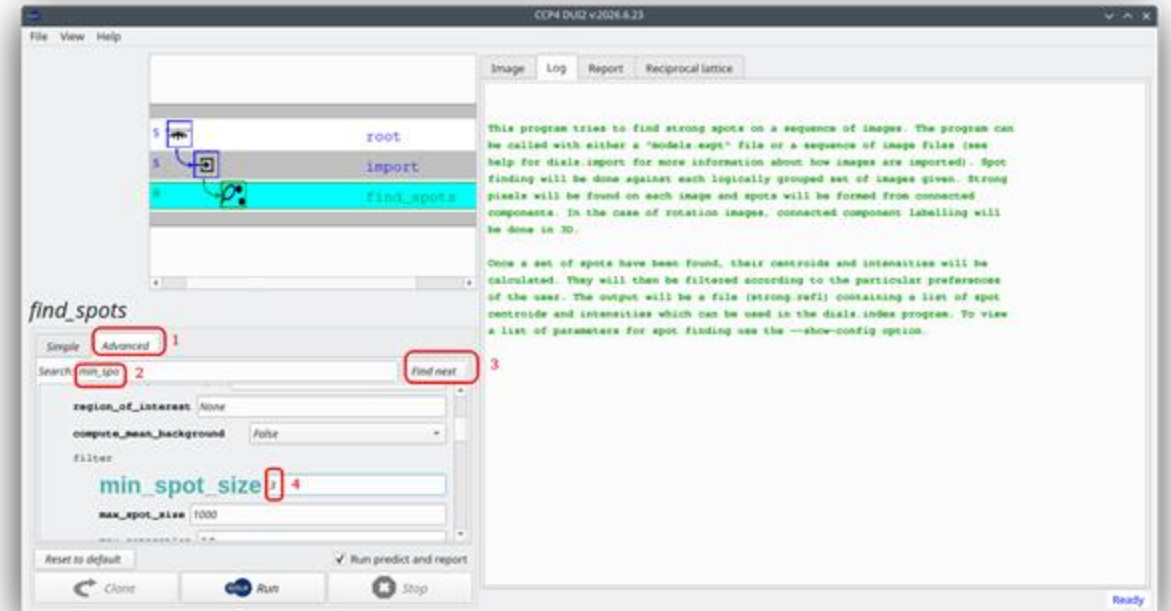
CHAPTER 1 · DISCOVER

Step 2

Find spots

- Select "find spots" from the left-hand panel.
- Under the Advanced tab, use the search box to find and set min_spot_size → 3.

This removes very small pixel islands that are not genuine Bragg spots. Click Run — the log reports a large number of spots found across 100 images.



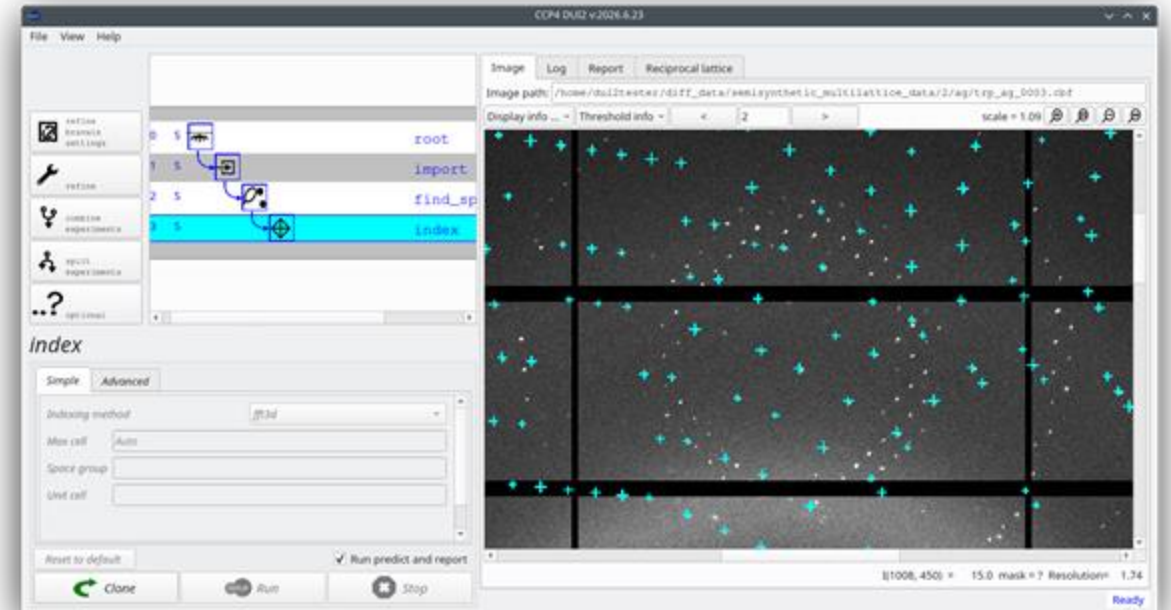
Setting min_spot_size in the Advanced tab

CHAPTER 1 · DISCOVER

Step 3

Index — single lattice first

- Select "index" and click Run with default parameters. Evidence for a second lattice shows up across the visualisation tabs — starting with the Image tab:



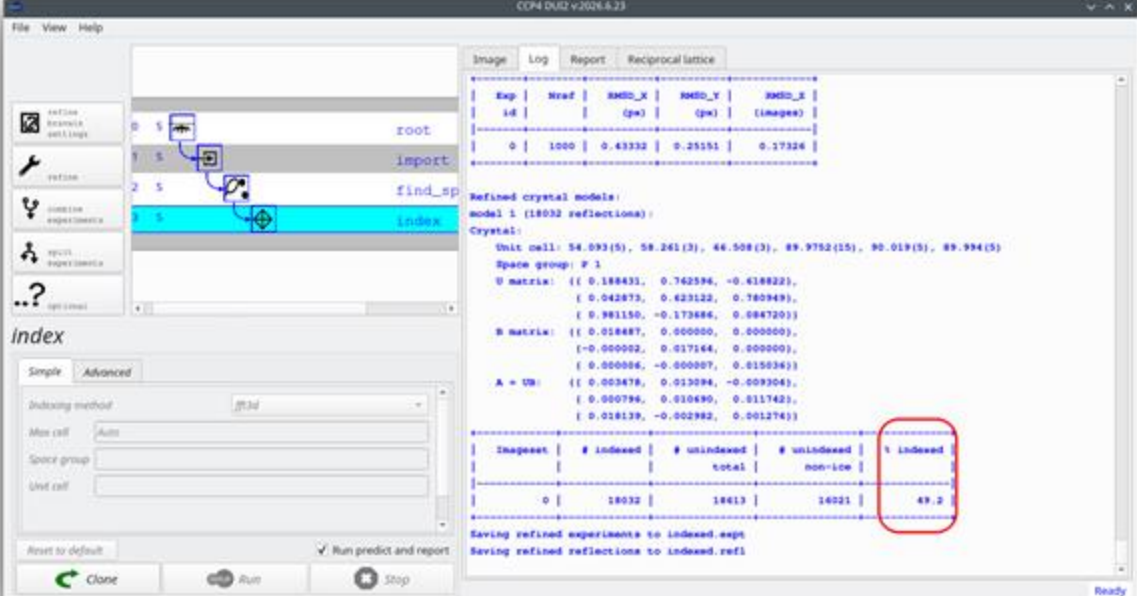
"Image" tab — predictions match one orientation; a systematic set of strong spots remains unpredicted.

CHAPTER 1 · DISCOVER

Step 3 (continued)

Index — single lattice first

- The Log tab tells the same story:



Refined crystal models:
model 1 (18032 reflections):

Crystal:

Unit cell: 54.093(5), 58.241(3), 46.508(3), 89.9752(15), 90.019(5), 89.994(5)

Space group: P 1

U matrix: $\begin{pmatrix} 0.188431 & 0.742594 & -0.418822 \\ 0.042873 & 0.423122 & 0.760949 \\ 0.981150 & -0.173884 & 0.584720 \end{pmatrix}$

B matrix: $\begin{pmatrix} 0.018487 & 0.000000 & 0.000000 \\ -0.000002 & 0.017164 & 0.000000 \\ 0.000004 & -0.000007 & 0.015034 \end{pmatrix}$

A = UB: $\begin{pmatrix} 0.003478 & 0.013094 & -0.009304 \\ 0.000794 & 0.010490 & 0.011742 \\ 0.018139 & -0.002982 & 0.001274 \end{pmatrix}$

Imageset	# indexed	# unindexed	# unindexed	% indexed
		total	non-ice	
0	18032	18613	14021	49.2

Saving refined experiments to indexed.expt
Saving refined reflections to indexed.refl

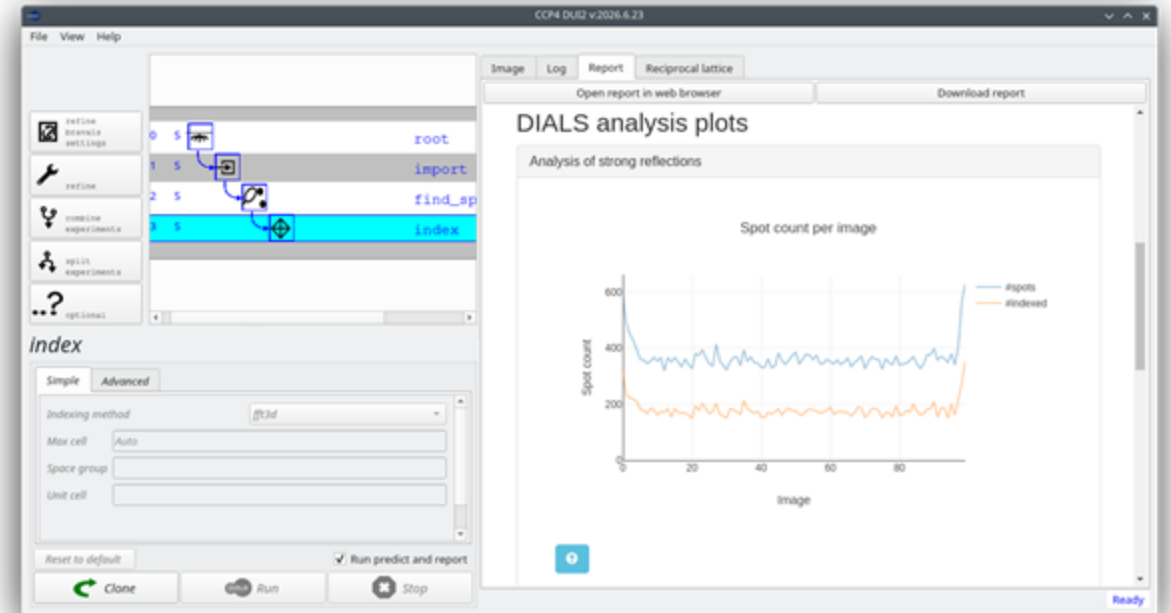
"Log" tab — the fraction of indexed spots is noticeably lower than the 80–95% expected for a clean dataset.

CHAPTER 1 · DISCOVER

Step 3 (continued)

Index — single lattice first

- The remaining two tabs make the second lattice unmistakable — starting with the Report tab:



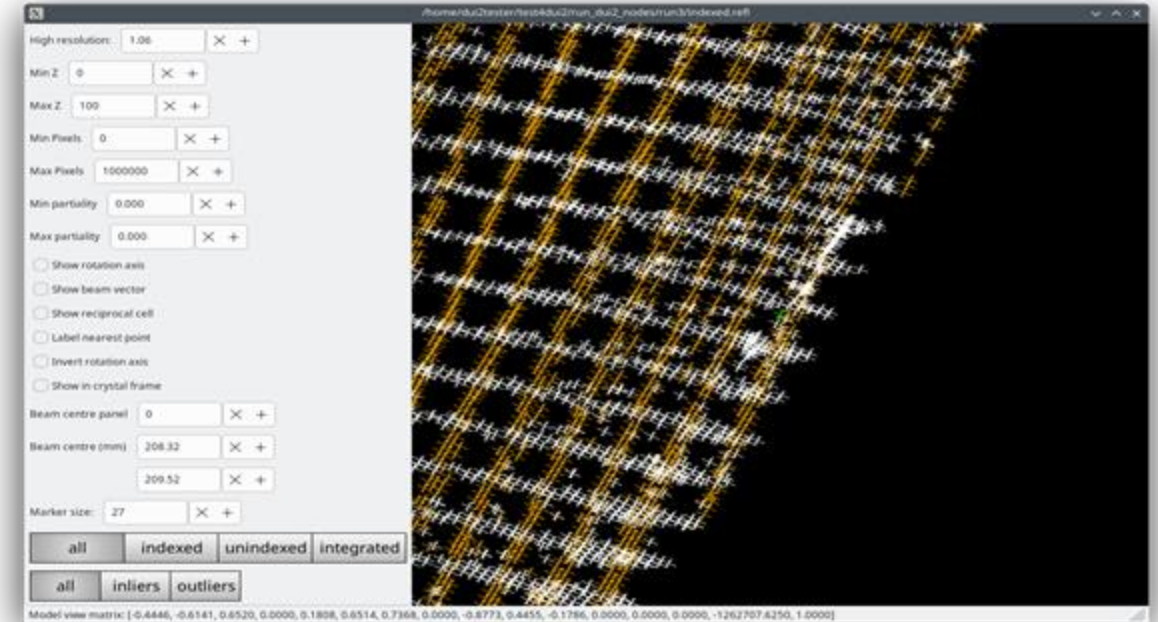
"Report" tab — Spot count per image: only around half the found spots appear as indexed.

CHAPTER 1 · DISCOVER

Step 3 (continued)

Index — single lattice first

- The Reciprocal lattice tab confirms it:



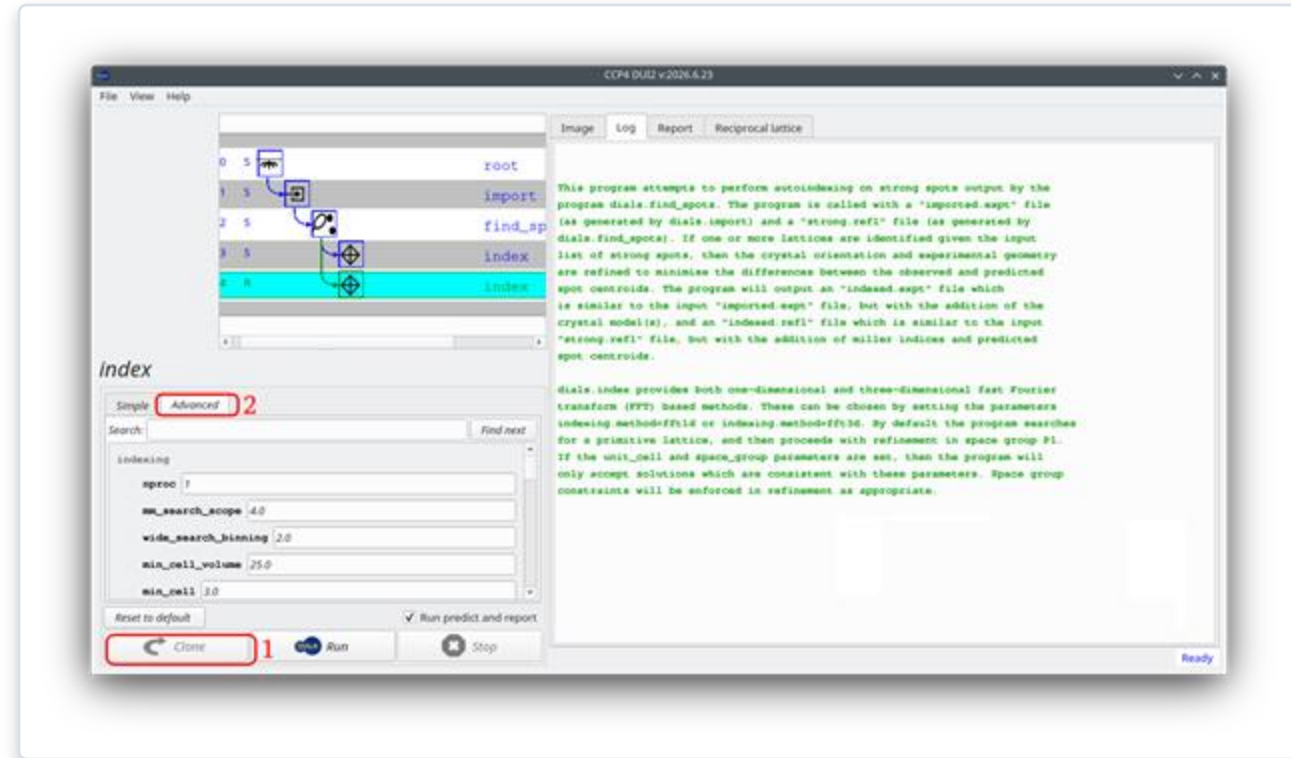
"Reciprocal lattice" tab — unindexed spots form their own ordered rows: the signature of a second lattice.

CHAPTER 1 · DISCOVER

Step 4

Clone and *Index* again, but with two lattices

- With the index node selected, click "Clone" — this creates an independent copy branching from the same find-spots input; the original single-lattice node is preserved as a reference.



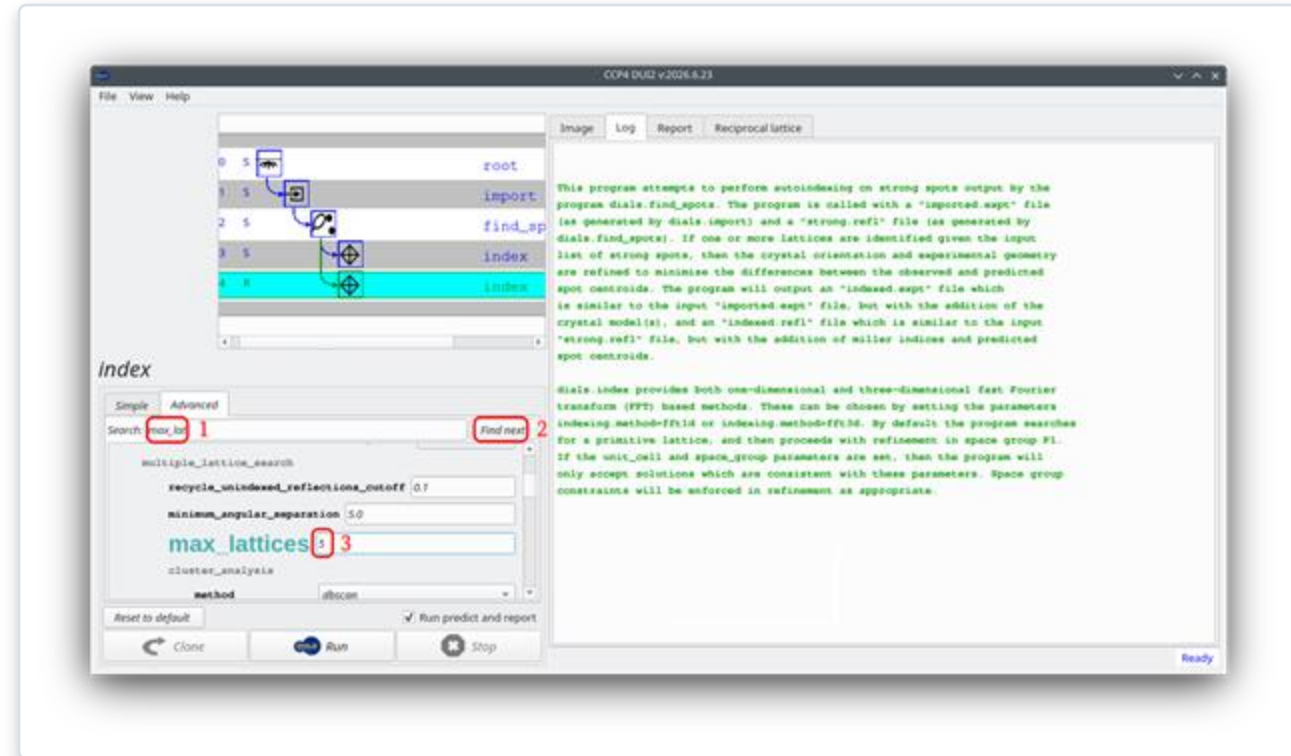
Cloning the index node

CHAPTER 1 · DISCOVER

Step 4 (continued)

Clone and *Index* again, but with two lattices

- In the cloned node's Advanced tab, set `max_lattices` → 5, then click Run. DIALS searches for up to five lattices but stops once no more can be justified — here it finds exactly two, accounting for ~95% of all strong spots.



Setting max_lattices in the Advanced tab

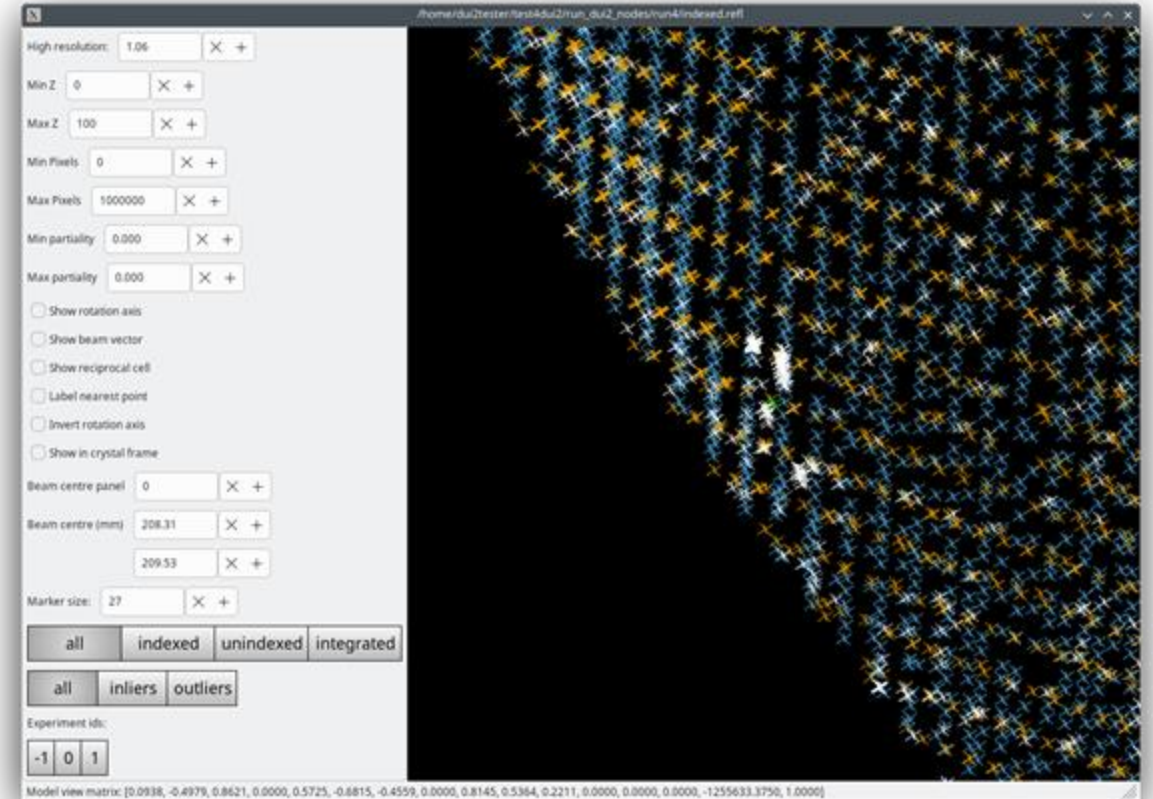
CHAPTER 1 · DISCOVER

Step 4

Result of new index step

- After re-indexing with max_lattices=5, the log reports two refined crystal models.
- Together they account for roughly 95% of all strong spots.

Two clearly distinct orientation families, each forming clean rows, with very few unindexed spots remaining.



Reciprocal lattice viewer after indexing with two lattices

CHAPTER 1 · DISCOVER

Step 5

Split experiments

- Select "split experiments" and click Run. DIALS creates one independent child node per lattice, each showing only its own reflections when you open its Image tab.
- Decision point: if a lattice fits poorly throughout the scan, there is no obligation to process it further. For this dataset, both branches look good.

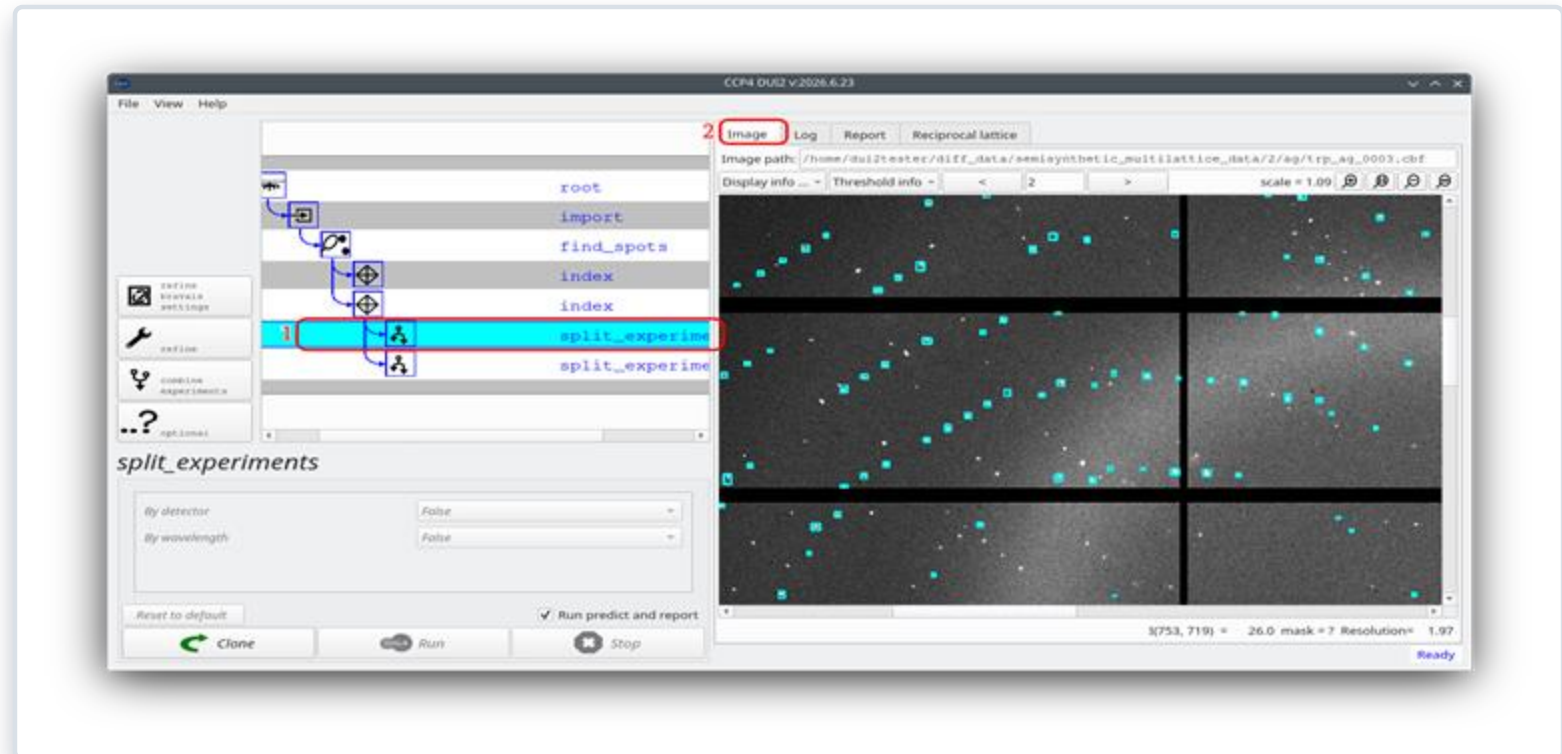


Image view — branch 0 (lattice 0)

CHAPTER 1 · DISCOVER

Step 5 (continued)

Split experiments

- Select "split experiments" and click Run. DIALS creates one independent child node per lattice, each showing only its own reflections when you open its Image tab.
- Decision point: if a lattice fits poorly throughout the scan, there is no obligation to process it further. For this dataset, both branches look good.

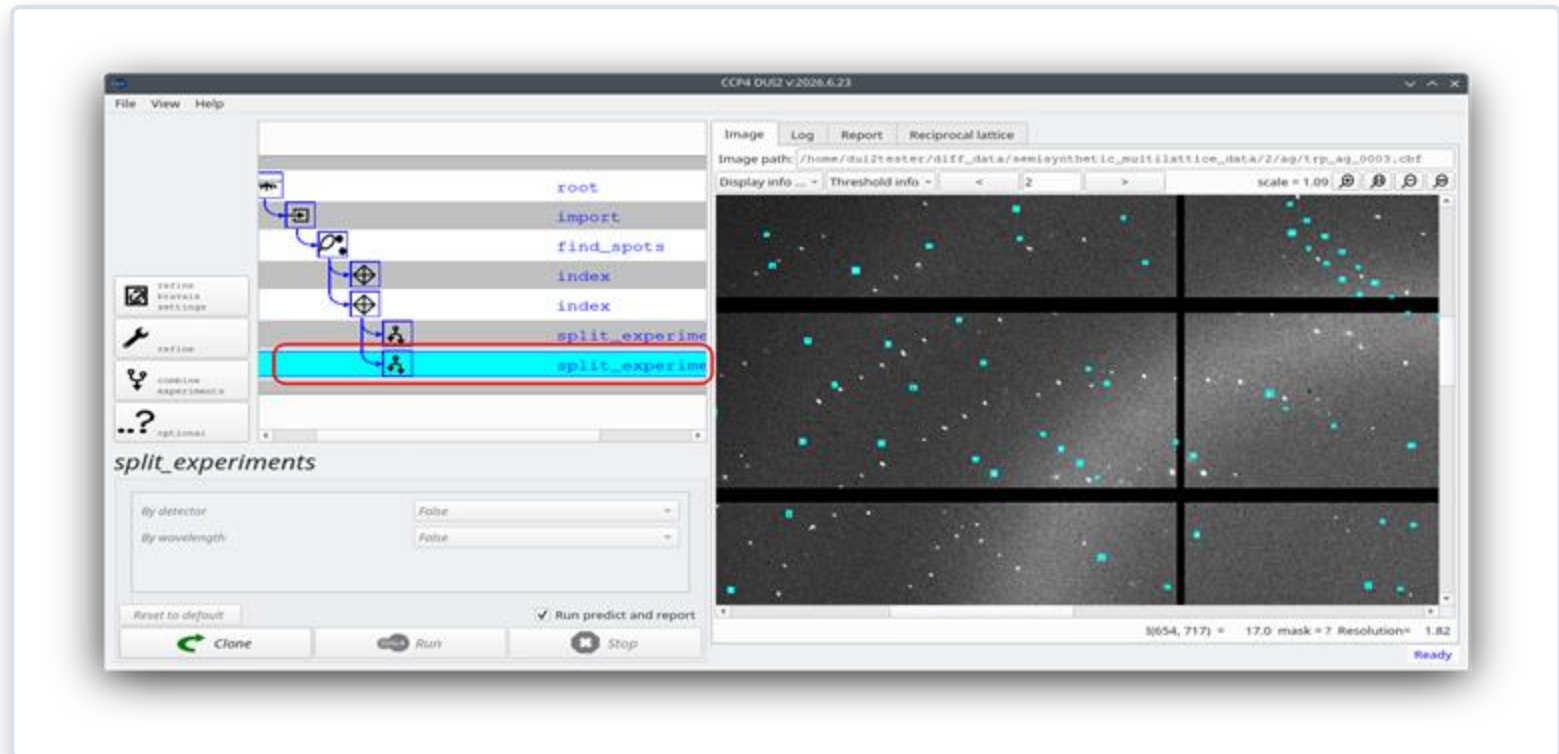


Image view — branch 1 (lattice 1)

CHAPTER 2

Process

From here the two branches are independent. Work through Bravais settings, refinement, and integration on each in turn.

Refine Bravais settings · Reindex · Refine · Integrate

Step 6

Refine Bravais settings

With a branch node selected in the history tree, select "refine bravais settings" and click Run.

A table of candidate Bravais lattice solutions is written to the log.

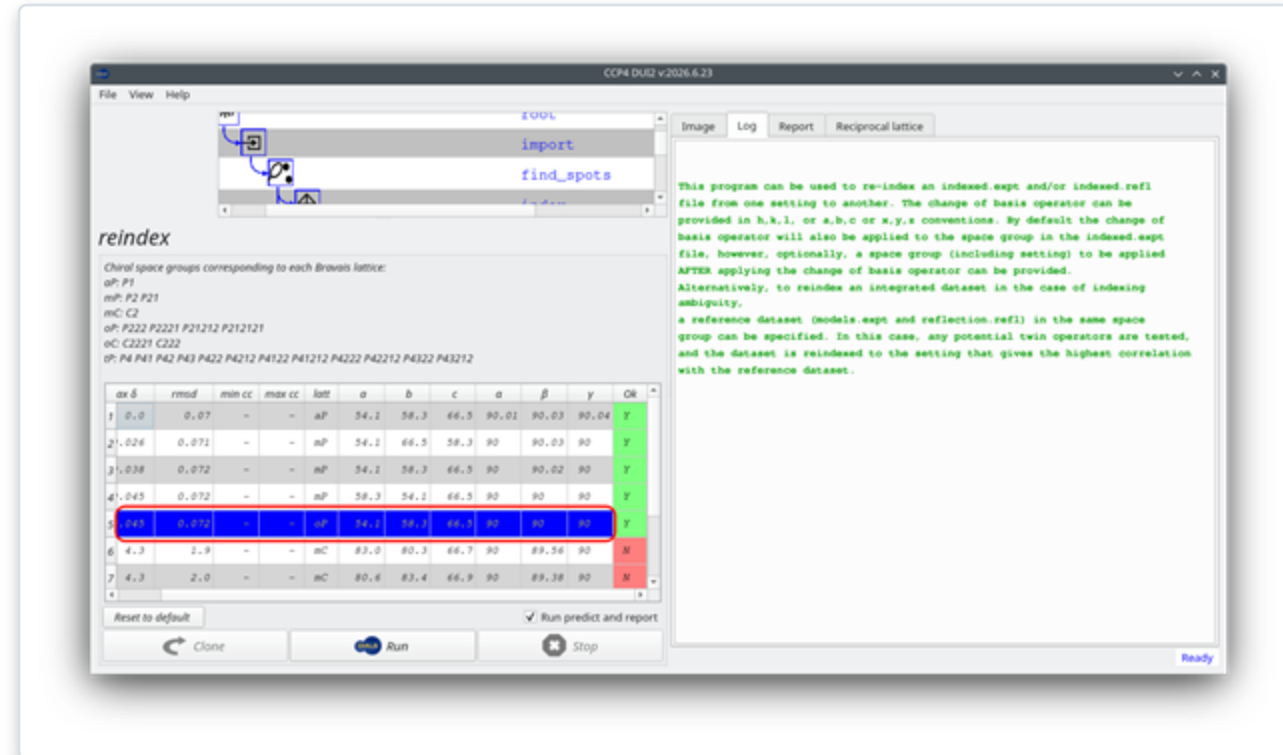
Repeat on branch 0, then branch 1 — the steps in this chapter are identical for both.

CHAPTER 2 · PROCESS

Step 7 Reindex

- Select "reindex". DUI2 displays an interactive table of the solutions from the previous step.
- Click the row for solution 5 — primitive orthorhombic (oP, P 2 2 2) — and DUI2 runs dials.reindex automatically.

The chosen solution should have a metric fit significantly closer to zero than the others, and be marked acceptable. If the highest-symmetry acceptable solution differs between branches, investigate before continuing.



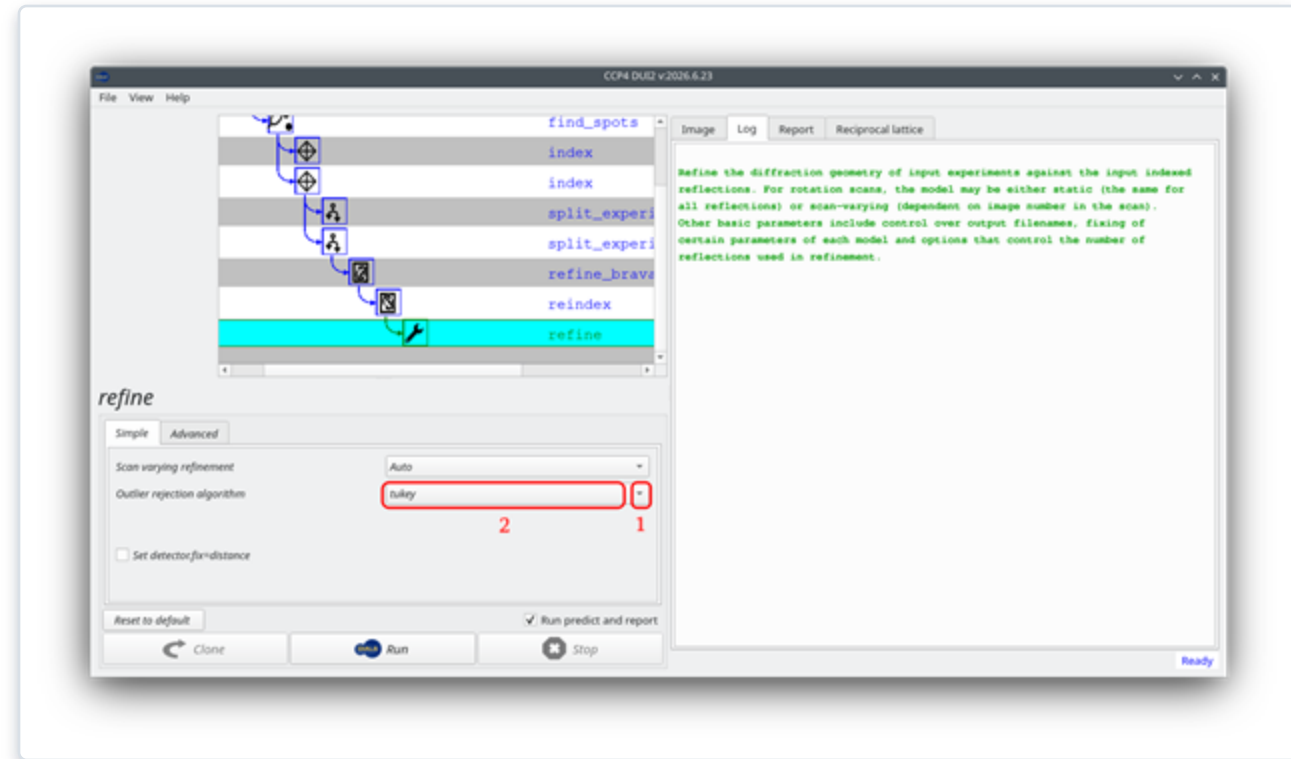
DUI2 about to run reindex

CHAPTER 2 · PROCESS

Step 8 Refine

- Select "refine". Under the Simple tab, set Outlier rejection algorithm → tukey.
- Click Run — scan-static and scan-varying refinement run for this lattice.

Check the Report tab: changes in cell parameters and orientation angles across the scan should look small and smooth.

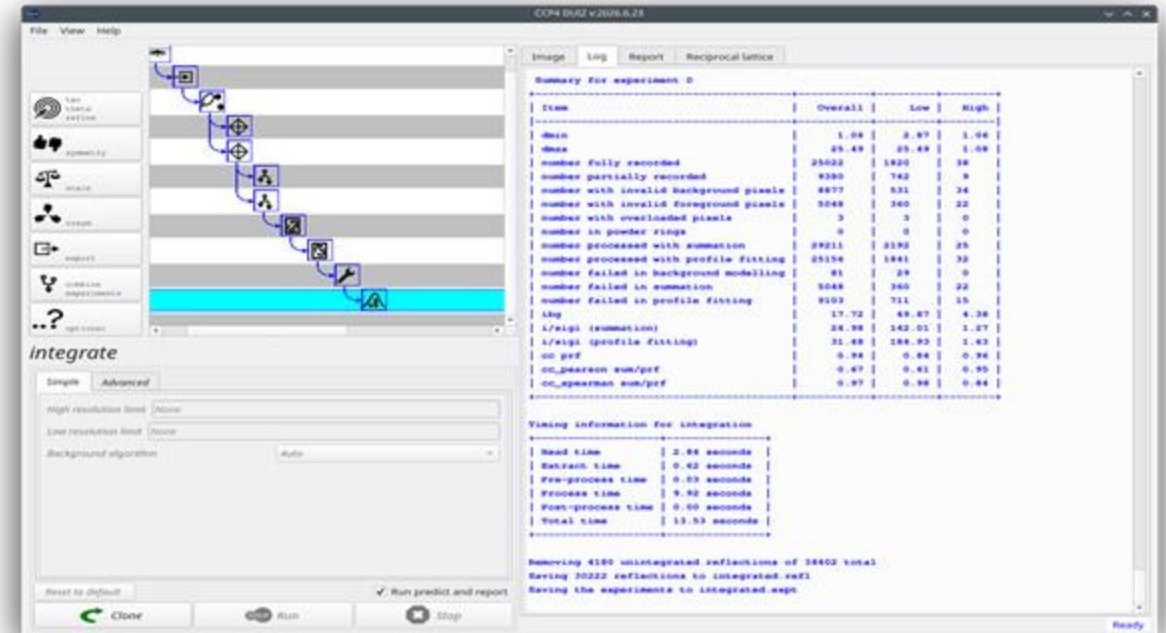


DUI2 about to run refine

CHAPTER 2 · PROCESS

Step 9 Integrate

- Select "integrate" and click Run with default parameters. Check the end of the log for "Summary for experiment 0" — $I/\sigma(I)$ and cc_{prf} should look reasonable.



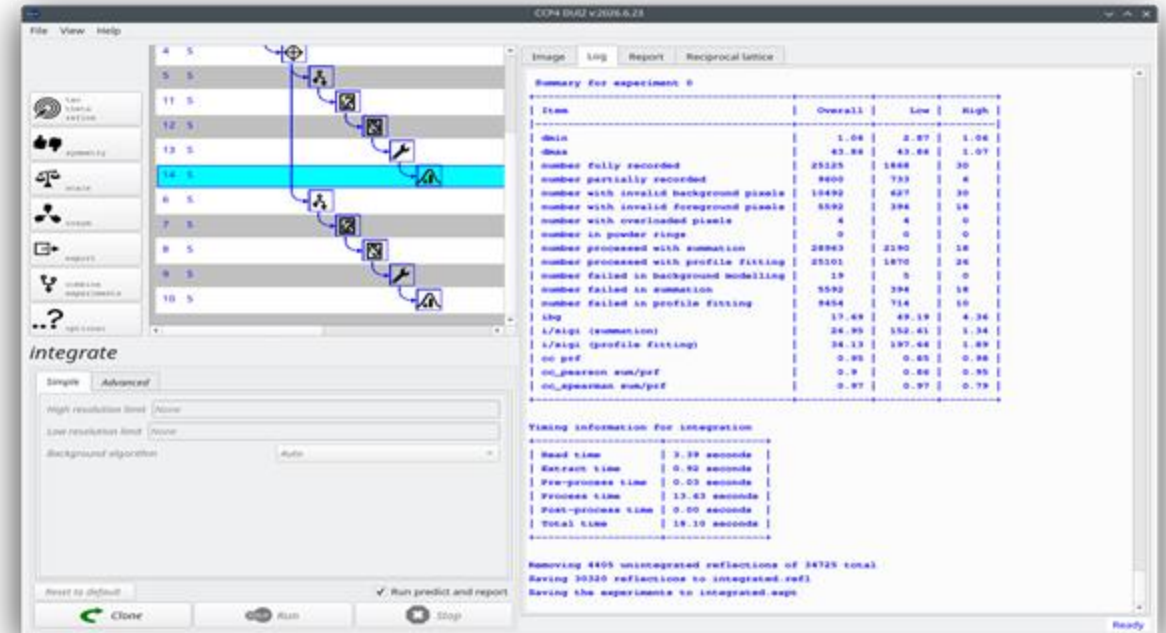
Log view after integrating one branch

CHAPTER 2 · PROCESS

Step 9 (continued)

Integrate

- Once steps 6–9 are complete on one branch, return to the split experiments node, select the other branch, and repeat.



Log view after integrating both branches

CHAPTER 3

Post integration

With both branches refined and integrated, run symmetry, scaling, and export independently on each branch.

Determine symmetry · Scale · Export

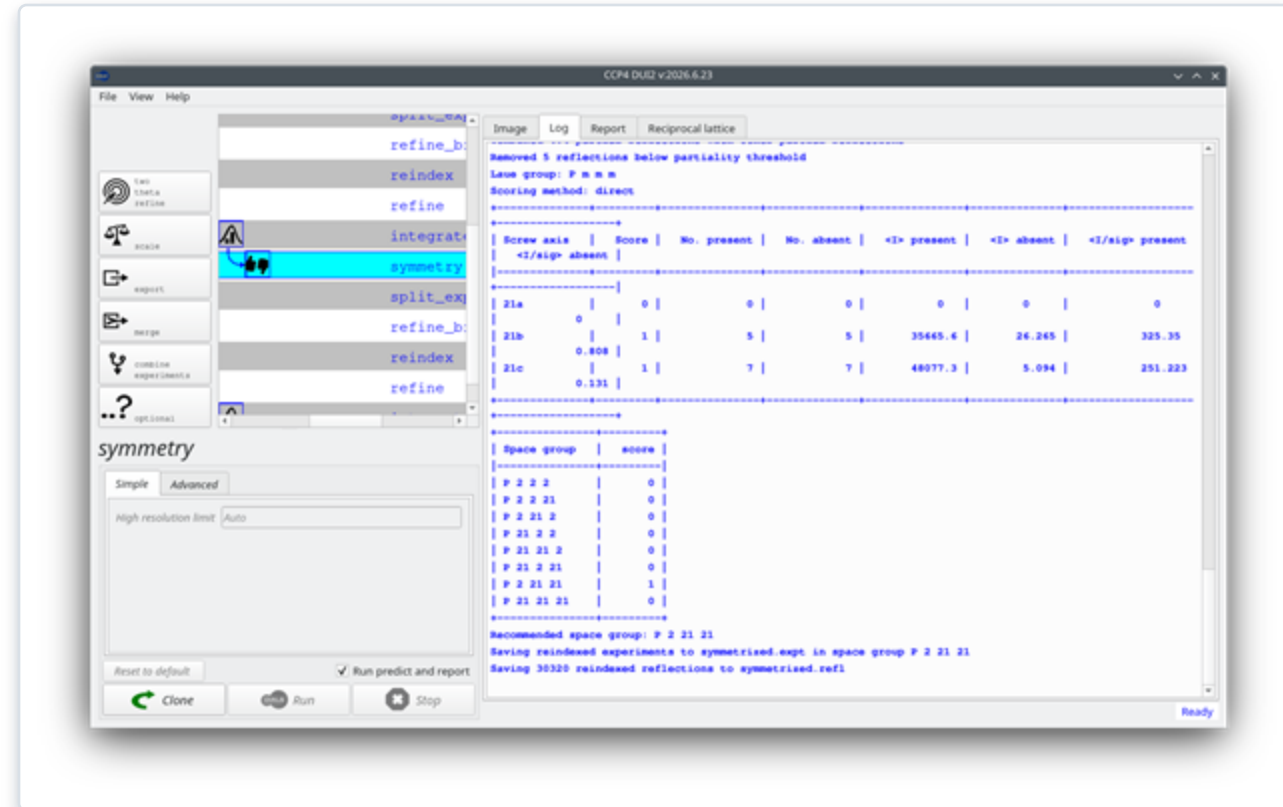
CHAPTER 3 · POST INTEGRATION

Step 10

Determine crystal symmetry

- Select "symmetry" and click Run.
- The analysis should return P m m m as the Laue group, with clear 2₁ screw axes indicating an orthorhombic space group (P 2₁ 2₁ 2₁ for the first dataset — axis labelling may differ for other archive datasets).

No reflections test the 2₁ screw along a for this dataset — worth testing both P 2₁ 2₁ 2₁ and P 2₁ 2₁ 2₁ during structure solution.



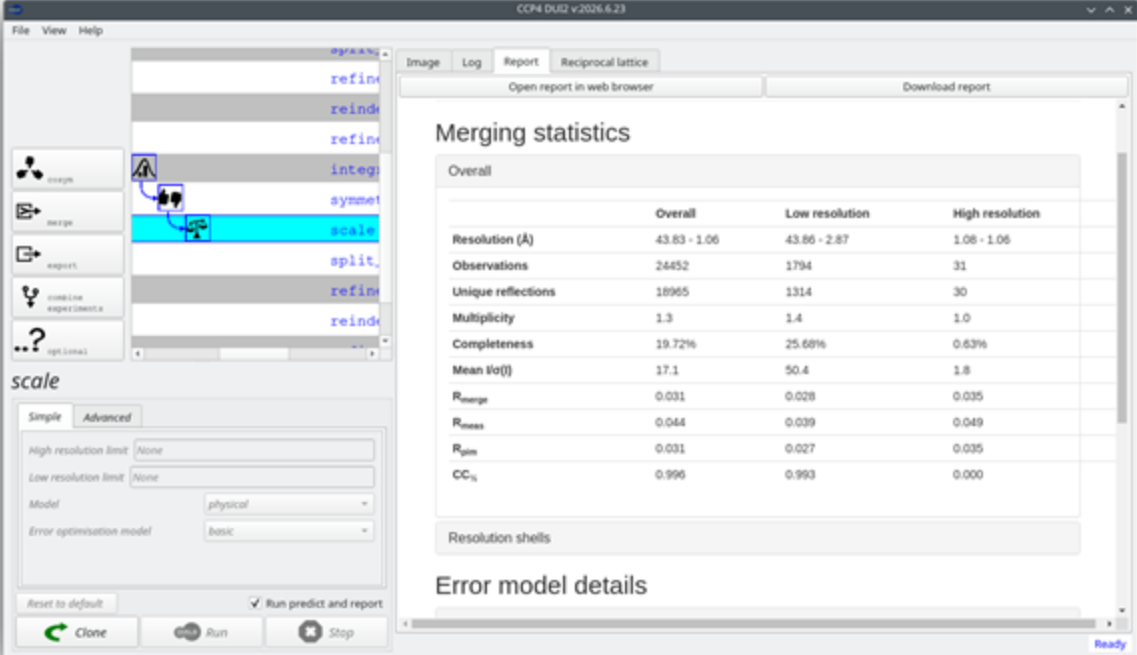
DUI2 with new symmetry applied

CHAPTER 3 · POST INTEGRATION

Step 11 Scale

- Select "scale" and click Run with default parameters.
- Check the Report tab for resolution-dependent statistics.

The data has not yet been merged at this stage — dials.scale reports merging statistics but retains all observations.



CPD DIALS v2026.6.23

Image Log Report Reciprocal lattice

Open report in web browser Download report

Merging statistics

	Overall	Low resolution	High resolution
Resolution (Å)	43.83 - 1.06	43.86 - 2.87	1.06 - 1.06
Observations	24452	1794	31
Unique reflections	18965	1314	30
Multiplicity	1.3	1.4	1.0
Completeness	19.72%	25.68%	0.63%
Mean I/σ(I)	17.1	50.4	1.8
R _{merge}	0.031	0.028	0.035
R _{meas}	0.044	0.039	0.049
R _{pim}	0.031	0.027	0.035
CC _{1/2}	0.996	0.993	0.000

Resolution shells

Error model details

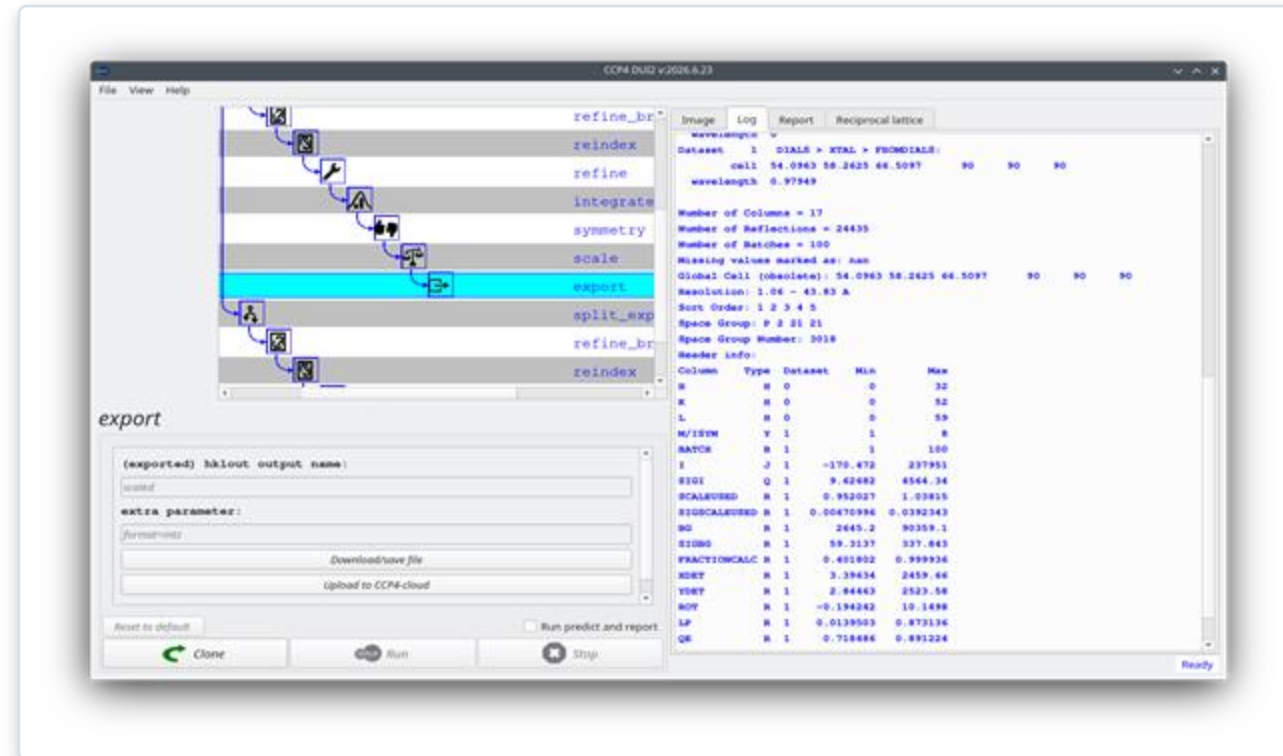
Ready

Report tab after scaling

CHAPTER 3 · POST INTEGRATION

Step 12 Export

- Select "export", click Run, then "Download/save file" to write the MTZ for this lattice. Repeat on the second branch for a separate MTZ per lattice.



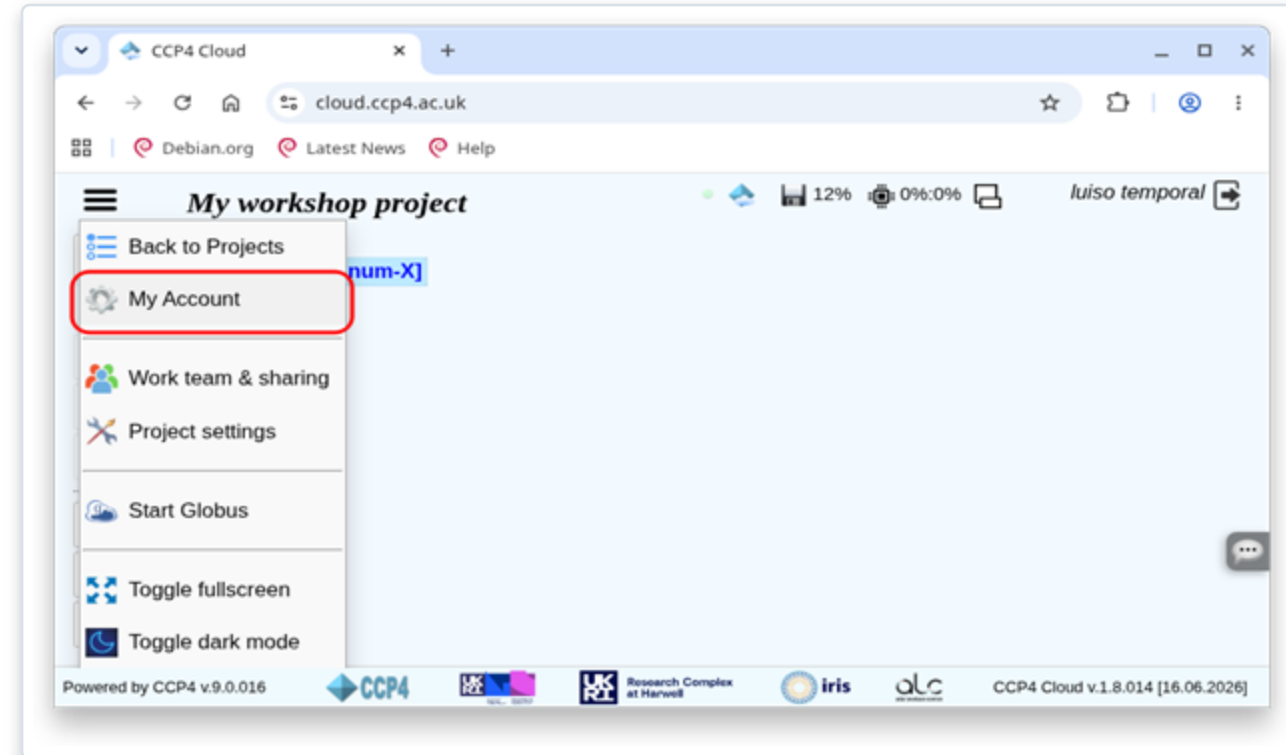
Export and download/save MTZ

CHAPTER 3 · POST INTEGRATION

Step 12 (continued)

Export

- To upload directly to CCP4 Cloud: note your project ID and "CloudRun Id" from My Account, then press "Upload to CCP4-cloud" in DUI2 and fill in the dialog — this creates a cloudrun node.



CCP4 Cloud project page

Summary

Discover

Single-lattice indexing exposed a second lattice via the Image, Log, Report, and Reciprocal lattice tabs; re-indexing with `max_lattices=5` and splitting produced two independent branches.

Process

Each branch moved through Bravais settings, reindexing to the shared oP solution, refinement with Tukey outlier rejection, and integration.

Post integration

Each branch was independently assigned symmetry (P m m m Laue group), scaled, and exported to its own MTZ — optionally uploaded to CCP4 Cloud.

Thank you

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<https://github.com/ccp4/DUI2/wiki>

<https://github.com/ccp4/DUI2/issues>

